EGL211 Project Report

Member Name	Admin No.
HOLY ANGELIN VIRSA IRWANTO	244876B
MIN THIHA	241450U
TAN YIN SIAH OLLEVYA EVALYN	244814Q
YADANAR HTUN	241451G
AWA ZILD POOT	243234R
BRANDON LIM JINHUI	232668Z

Executive Summary

This technical report aims to document the design and implementation of a highly available 3-tier web application architecture on Amazon Web Services (AWS). This project will also show the testing that was performed to ensure that all requirements of the project were met. This project will showcase core cloud computing skills and concepts which includes but is not limited to network isolation, load balancing, auto scaling, highly available databases, and the best practices to ensure high security.

Some of the key parts of this project that was implemented includes:

- The creation of a custom Virtual Private Cloud (VPC) with 6 subnets that span across 2 Availability Zones (AZs).
- Application Load Balancer (ALB) which is for distribution of traffic.
- Auto Scaling Group (ASG) to maintain having 2 application Amazon Elastic Cloud Compute (EC2) instances running at all times.
- Implement a Multi-AZ Relational Database Service (RDS) MySQL database.
- Launch a Bastion Host EC2 instance for secure administrative access.
- Create Network Address Translation (NAT) Gateways (1 for each AZ) to ensure private subnets can access the internet.
- Creation of Security Groups and Network Access Control Lists (NACLs) for defence in depth
- Elastic Block Store (EBS) Snapshots for business continuity
- CloudWatch alarms to monitor the status of the web app instances.

Contents

Executive Summary 2
Table of Responsibilities 5
Introduction 6
- Background 6
- Objectives 6

Implementation and Discussion 6
- VPC and Networking (Requirement 4.1) 6
  - VPC Creation 6
  - Subnet Configuration 7
  - Internet Gateway 8
  - NAT Gateways 11
  - Route Tables 12
  - Enable Auto-assign Public IP 14
- Security Groups (SGs) 14
- Network ACLs 15
- Baston Host Configuration 16
  - Connecting via SSH 17
  - SSH to Private Instances via Bastion in PuTTY 18
- Application Tier 18
  - Launch Template 18
  - Target Group 21
- Database Tier - RDS MySQL 25
  - DB Subnet Group 25
  - RDS MySQL Instance 26
  - Testing Database Connectivity 27
- S3 Storage Configuration 28
- Production Recommendation 31
- CloudWatch Monitoring 32
- Testing and Validation 33
  - Web Application Access 33
  - High Availability Test 34
  - Auto Scaling Self-Healing 34
  - Bastion SSH Access 36
  - Database Connectivity 36
  - Database Not Publicly Accessible 37

Private Instance Internet Access 37
S3 Testing 37
Problem & Challenges 39
IAM Role Creation Restrictions 39
EC2 Metadata Services (IMDSv2) 39
Bastion Host IP Change 39
Individual Reflections 40
Yadanar's Reflection 40
Ollevya's Reflection 40
Min's Reflection 41
Zilds's Reflection 41
Angelin's Reflection 42
Brandon's Reflection 42
Recommendations 43
Security Enhancements (AWS, 2024) 43
Cost Optimization (AWS,2024) 44
Operational Excellence (AWS, 2024) 44
Conclusions 44
References 45
Appendices 46

Table of Responsibilities

Member	Task Allocated	Report Contribution	Remarks
MIN THIHA	Implement application tier scalability and high availability		
YADANAR HTUN	Task 4.2	Completed 4.2 - Database Tier High Availability (RDS Multi-AZ deployment, subnet group configuration, security group setup, and connectivity testing)	Task 4.2 - 90% And supported overall system integration and troubleshooting
TAN YIN SIAH OLLEVYA EVALYN	Task 4, 6 and 7	Completed 4.3 - deploy bastion host in public subnet, ensure connectivity from private instance auto created from load balancing to database. Screenshotting configurations and test results, individual reflection.	Task 4 - 17% Task 6 - 18% Task 7 - 17%
AWA ZILD POOT	Task 4.4	Helped implement the database by trouble shooting why it can't be deployed properly (Fixed by reducing allocated storage and type). Completed 4.4 by creating a snapshot of app instance in the private subnet	Task 4.4 - 90% Alongside general trouble shooting for other tasks
BRANDON LIM JINHUI	Task 4.5	I was mainly responsible for setting up the S3 bucket, configuring the bucket policy, and testing file upload and download using curl and preassigned URLs. I also helped troubleshoot permission issues and ensured the storage setup met the project security requirements.	Task 4.5 - 90% Task 6 - 15%
HOLY ANGELIN VIRSA IRWANTO	I contributed in all the tasks.	I was mainly responsible for report writing and preparing the account for the demonstration. I had to ensure the report was cohesive	All Tasks at least 20% per task

Introduction

Background

Cloud computing heavily transformed how IT infrastructures are deployed (Rance, 2024). Through AWS, companies are now able to implement scalable, highly-available and cost-effective solutions. In this project, we deploy a 3-tier architecture which separates applications into application, presentation and data layers which will allow for independent scaling for improvements and efficiencies and improved security.

Objectives

In this project, we create a custom VPC with proper network segments across multiple AZs and implement proper routing with Internet Gateway (IGW) and NAT Gateways. We will also be deploying an ALB for proper traffic distribution and configure an ASG for automated management of instances.
We will also need to implement a multi-AZ RDS MySQL to ensure high availability for the database and deploy a Bastion Host for secure administrative access. We will also ensure business continuity by having Snapshots and monitoring through CloudWatch. We also ensure that we can configure S3 Buckets access through private instances without needing stored credentials.

This project will be implemented in 2 regions, AWS US East (N. Virginia) us-east-1a and us-east-1b AZs, and in these AZs will include the network, compute, database, storage and security configurations.

Implementation and Discussion

VPC and Networking (Requirement 4.1)

VPC Creation

Purpose: VPCs provide isolated network environments which ensures that the company has full control of IP addressing, subnet and routing configurations.

Configuration:

- VPC Name: proj-cas-VPC
- CIDR Block: 10.0.0.0/16 (provide 65,536 IP addresses)
- Region: US East (N. Virginia)

Procedure:

1. Create VPC → Select “VPC Only” and name the VPC “proj-cas”
2. Enter 10.0.0.0/16 as IPv4 CIDR block
3. Leave IPv6 “No IPv6 CIDR block”
4. Set the Tenancy to “Default” and click “Create VPC”

Making our IPv4 CIDR block /16 provides us with more than enough IP addresses for scalability in case of expansions. This CIDR block also ensures that we are in 1918 private address space. This ensures no conflicts with real public website’s IP, security (Gridelli, 2025).

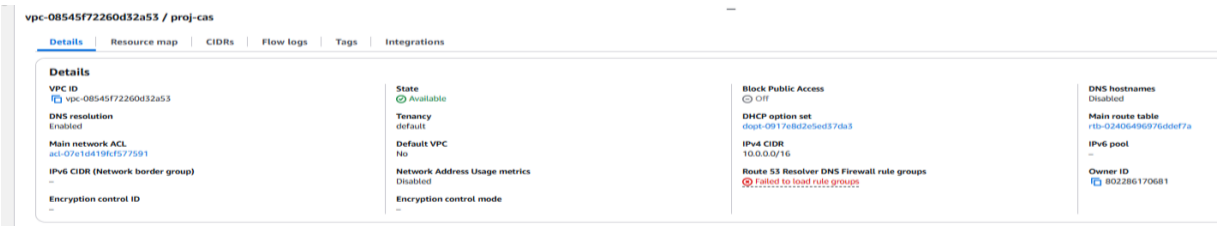


Figure 1: VPC Configuration Details

Subnet Configuration

Purpose: Subnets enable the VPC to be divided into smaller network segments. In this project, we created a total of 6 subnets, 3 in each AZ, for high availability (Senarath, 2025).

Subnet Name	CIDR Block	AZ	Type	Purpose
pub-sub-1a	10.0.1.0/24	us-east-1a	Public	ALB, Bastion, NAT GW
pub-sub-1b	10.0.2.0/24	us-east-1b	Public	ALB, NAT GW
priv-app-sub-1a	10.0.10.0/24	us-east-1a	Private	App Instances
priv-app-sub-	10.0.20.0/24	us-east-1b	Private	App Instances

1b				
priv-db-sub-1a	10.0.100.0/24	us-east-1a	Private	RDS Primary
priv-db-sub-1b	10.0.200.0/24	us-east-1b	Private	RDS Standby

Procedure:

1. Go to Subnets → Create Subnet
2. Select “proj-cas” VPC and enter the subnet name, choose the AZ, and enter the CIDR block.
3. Click “Create Subnet”
4. Repeat for all 6 subnets.

Rationale: Having subnets across multiple AZs ensures that if 1 AZ fails, it does not bring down all the applications and services. Having separate public and private subnets will help to enforce network isolation.

Internet Gateway

Purpose: An IGW allows communication between instances in the VPC and the internet for the public subnets in the VPC.

Procedure:

1. Internet Gateway → Create Internet Gateway
2. Name it “proj-igw” and click “Create Internet Gateway”
3. Select the IGW → go to “Actions” and choose “Attach to VPC”
4. Select “proj-cas” VPC and attach the IGW to it.

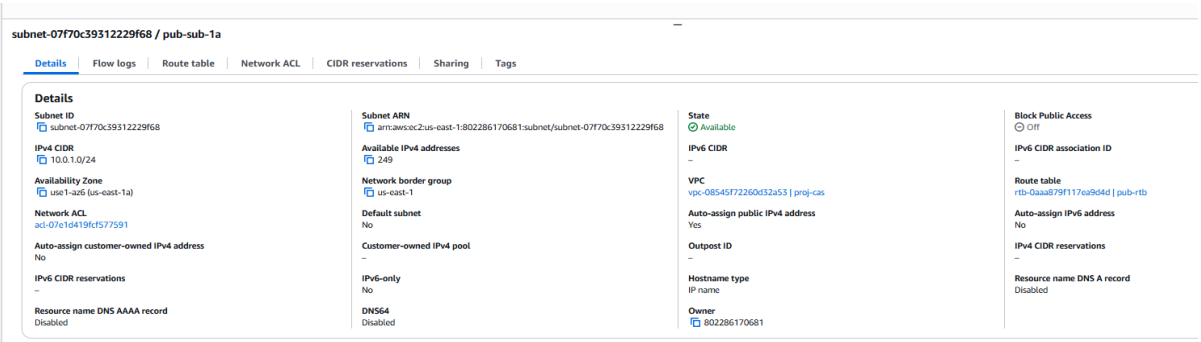


Figure 2:Configuration for Public Subnet in us-east-1a

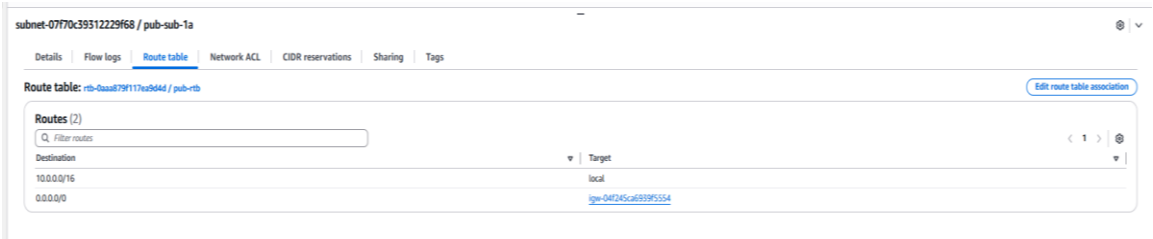


Figure 3:Shows the route table that it is associated with and the target IGW for pub-sub-1a

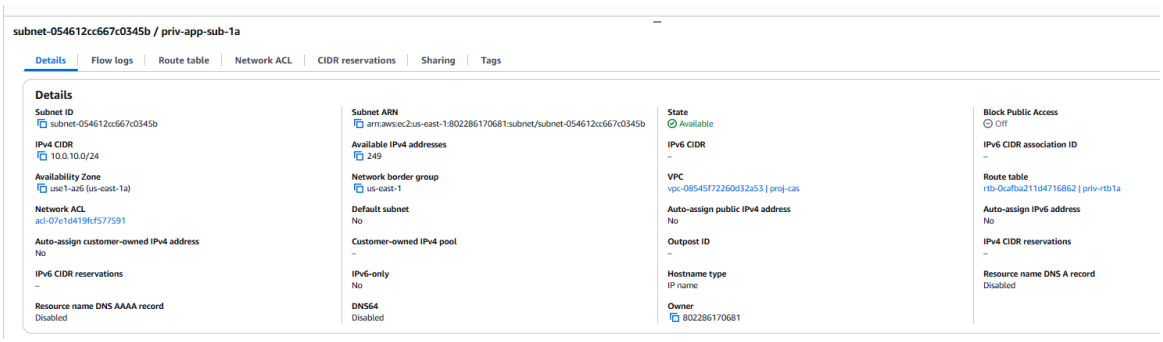


Figure 4: Configurations for private subnet in us-east-1a for Web Application

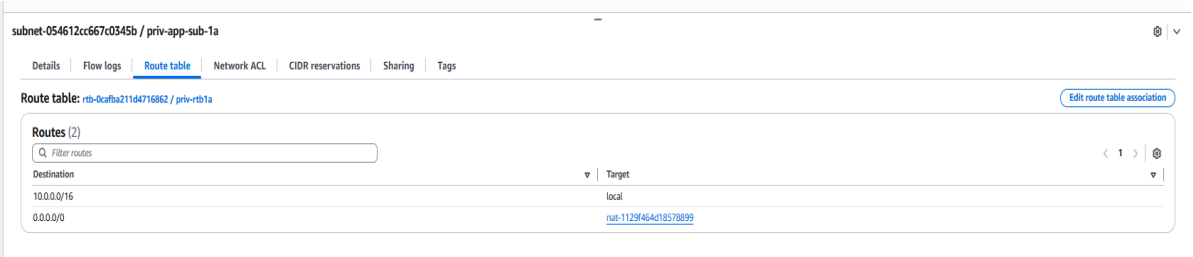


Figure 5: Shows the route table it is associated with and the target NAT Gateway for priv-app-sub-1a

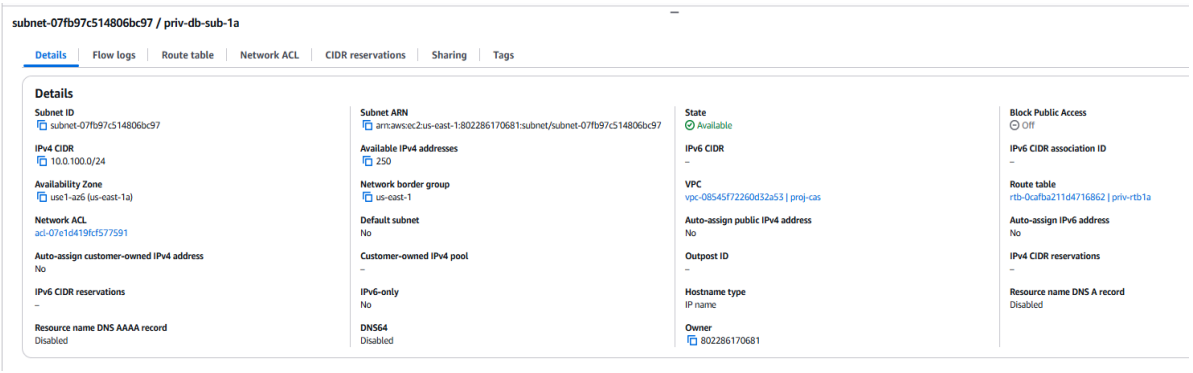


Figure 6: Configurations for Private Subnet in us-east-1a for Database

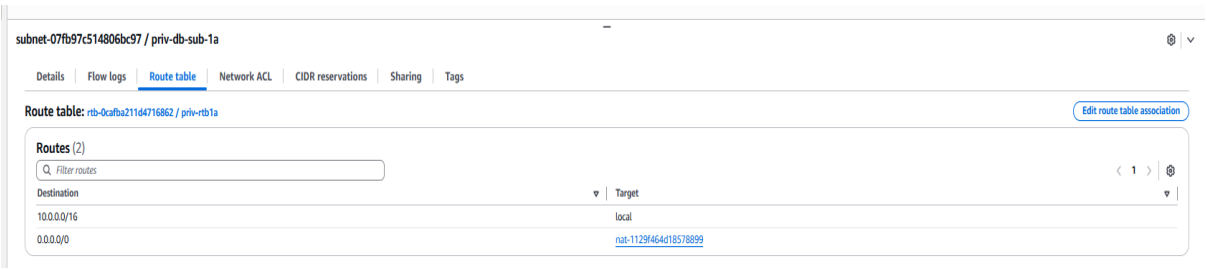


Figure 7: Shows the route table it is associated with and the target NAT Gateway for priv-db-sub-1a

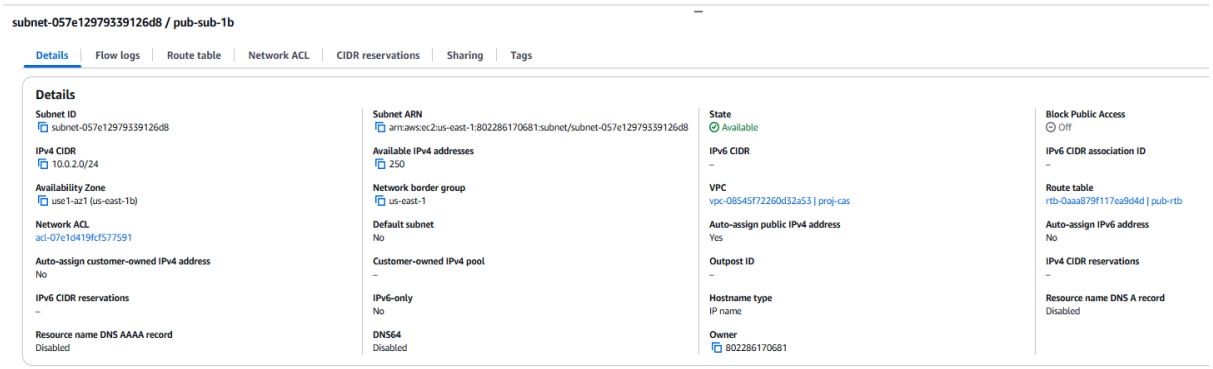


Figure 8: Configurations for Public Subnet in us-east-1b

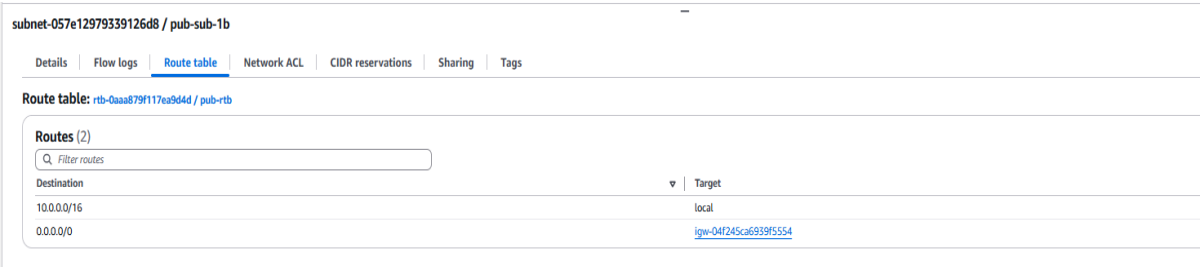


Figure 9: Shows the route table it is associated with and the target IGW for pub-sub-1b

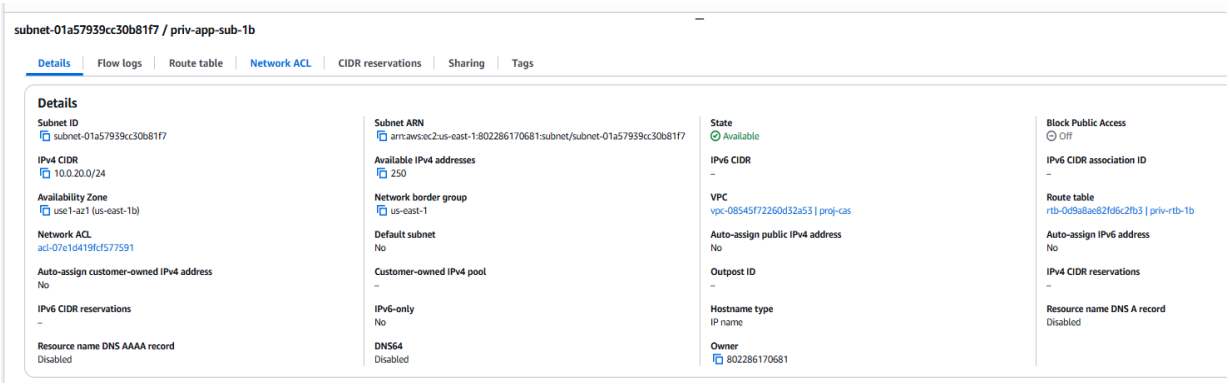


Figure 10: Configurations for Private Subnet in us-east-1b for Web Application

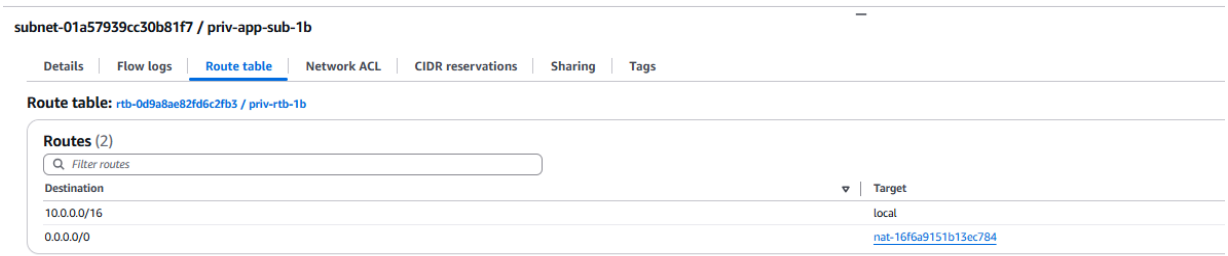


Figure 11: Shows route tables associated with the private subnet and the targeted NAT gateway

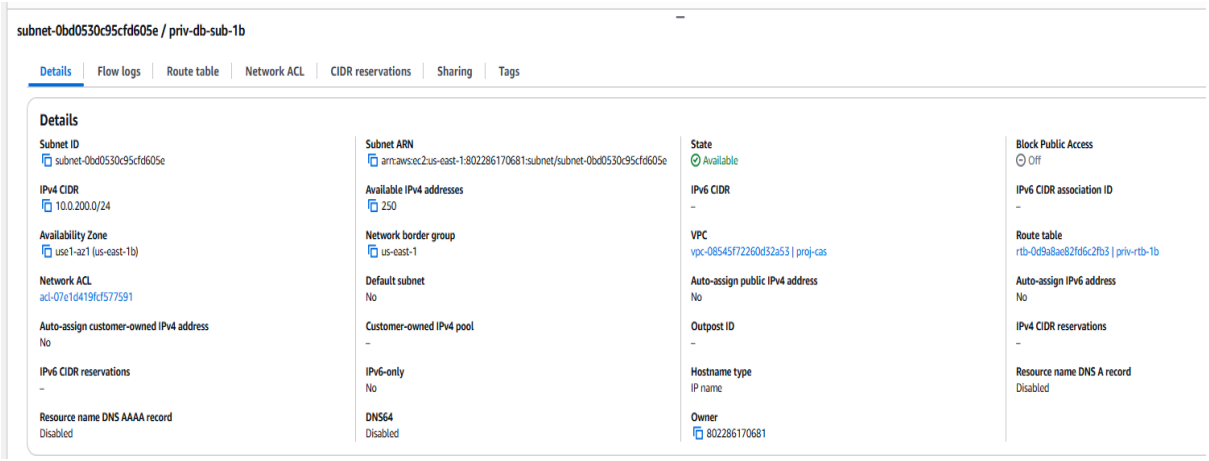


Figure 12: Configuration for Private Subnet in us-east-1b for Database

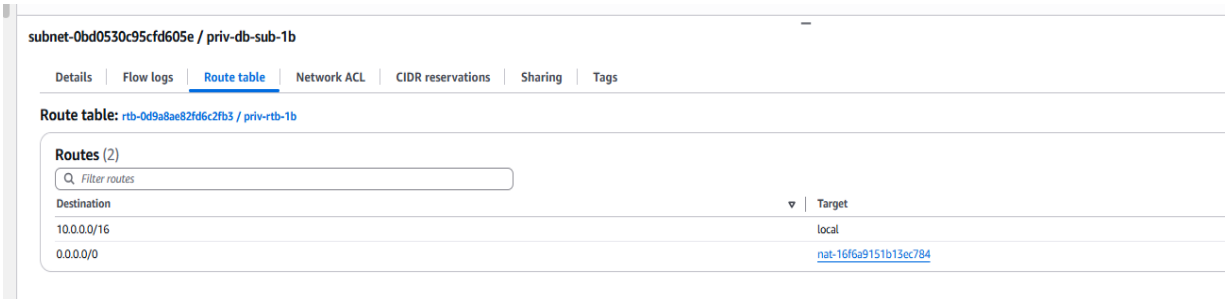


Figure 13: Shows the route tables associated with the subnet and the target NAT Gateway it is routed towards

NAT Gateways

Purpose: This will allow instances with private subnets to access the internet for patching and updating purposes while still being inaccessible from the internet.

Why did we make 2 NAT Gateways:

- If we have a single NAT Gateway, this would mean that if the NAT fails to connect to the internet, all of our private subnets will not be able to access the internet for patching and updates.
- Having 2 NAT gateways (one in each AZ) will allow for high availability for regular patching and updates.

Procedure:

1. NAT Gateways → Create NAT Gateway
2. Names: “proj-nat-1a-gw” and “proj-nat-1b-gw”
3. Subnet: “pub-sub-1a” and “pub-sub-1b” respectively
4. Connectivity type: “Public”
5. “Allocate Elastic IP” and then proceed to press “Create NAT Gateway”
6. Ensure that the status becomes “Available”

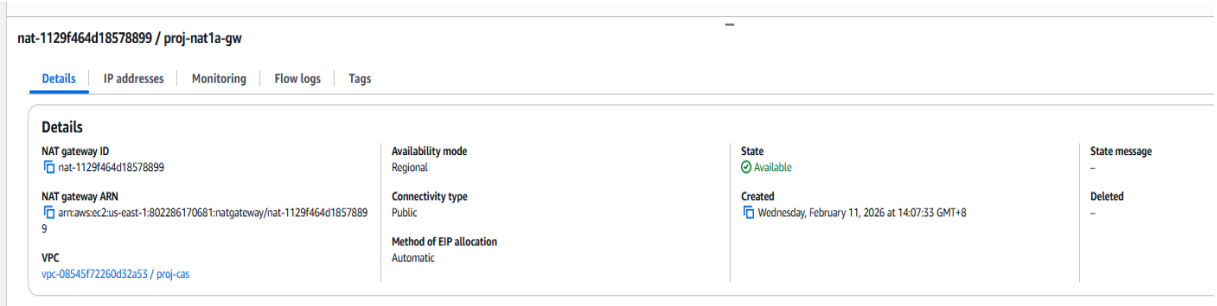


Figure 14: Shows the NAT Gateway Configurations for us-east-1a

Figure 15: Shows the subnets associated with the NAT Gateway

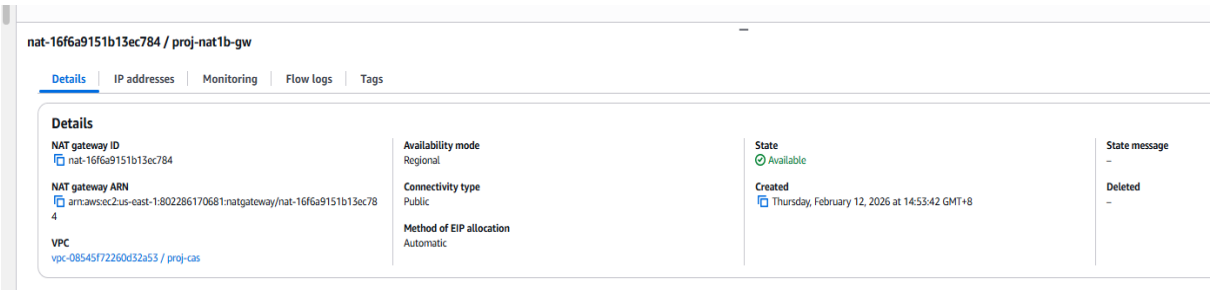


Figure 16: Shows the NAT Gateway Configurations for us-east-1b

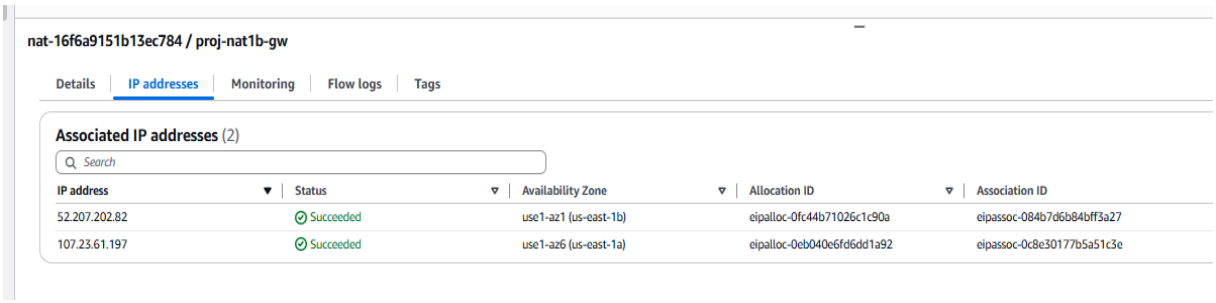


Figure 17: Shows the subnets Associated with the NAT Gateway

Route Tables

Purpose: Route tables will help to determine where the network traffic is being directed to.

Route Table	Destination	Target	Associated Subnets
pub-rtb	0.0.0.0/0	IGW	pub-sub-1a, pub-sub-1b
priv-rtb-1a	0.0.0.0/0	nat-gw-1a	priv-app-sub-1a, priv-db-sub-1a
priv-rtb-1b	0.0.0.0/0	nat-gw-1b	Priv-app-sub-1b, priv-db-sub-1b

Procedure for Public RTB:

1. Route Tables → Create Route Table
2. Name: “pub-rtb” and select “proj-cas” as the VPC and click “Create Route Table”
3. Select “pub-rtb” go to the Routes tab and edit routes
4. Add route: 0.0.0.0/0 and target it towards “proj-igw” and save changes
5. Subnet associations → edit and select both public subnets in the 2 AZs and save

Procedure for Private RTBs:

1. Create “priv-rtb-1a” and “priv-rtb-1b”, add a route to 0.0.0.0/0 and target them towards “nat-gw-1a” and “nat-gw-1b” respectively.
2. For priv-rtb-1a associate private subnets from us-east-1a
3. For priv-rtb-1b associate private subnets from us-east-1b

Rationale: Private subnets in their respective AZs are routed through their own NAT GW, so that if one fails, the subnets in the other AZs can still have internet access.

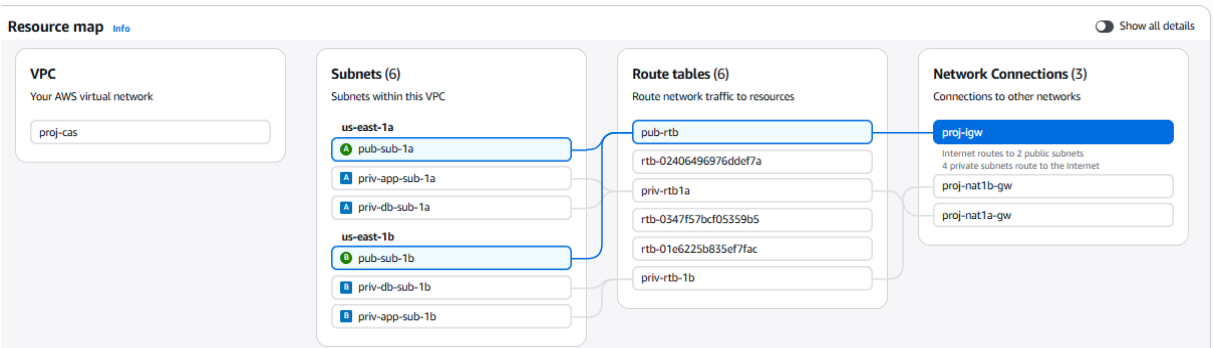


Figure 18: Resource map in VPC showing where public subnets routing towards proj-igw

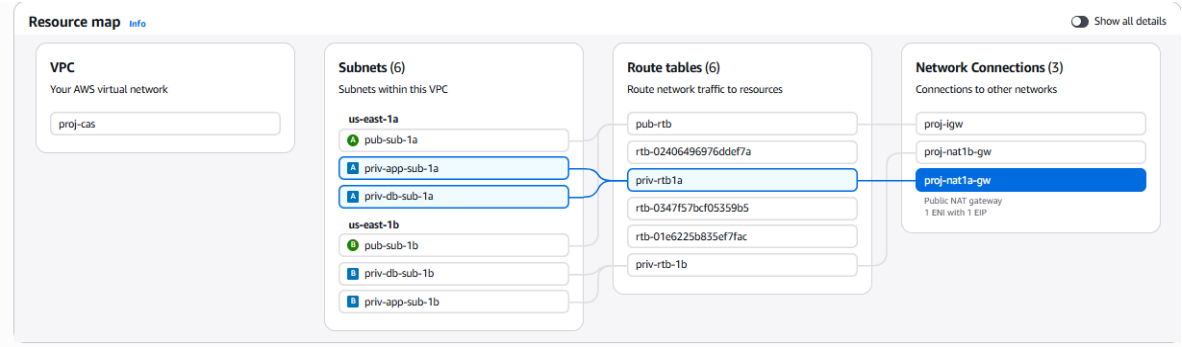


Figure 19: Resource map in VPC showing private subnets in us-east-1a route towards proj-nat-1a-gw

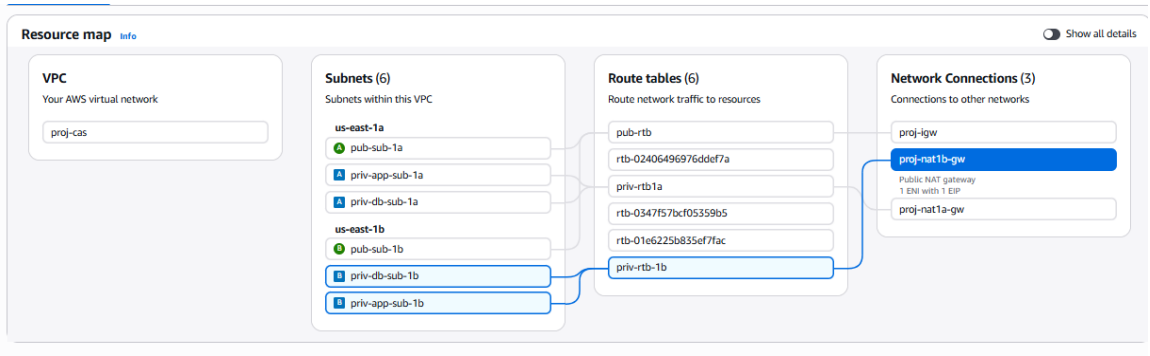


Figure 20: Resource map in VPC showing private subnets in us-east-1b routing towards proj-nat-1b-gw

Enable Auto-assign Public IP

Purpose: Instances in public subnets need public IPs for internet accessibility.

Procedure:

1. Go to Subnets and select “pub-sub-1a”
2. Click “Actions” and proceed to edit subnet settings
3. Check “Enable auto-assign public IPv4 address” and save
4. Repeat for “pub-sub-1b”

Security Groups (SGs)

Purpose: This acts as virtual firewalls at instance level and are stateful. This means that return traffic is automatically allowed. Outbound rules remain the same for all SGs, which allow traffic from anywhere at IPv4.

Security Group	Inbound Rules	Purpose
alb-sg	HTTP(80), HTTPS(443) from	Allow web traffic to ALB

	0.0.0.0/0	
bastion-sg	SSH(22) from 0.0.0.0/0	Allows SSH to Bastion only
app-sg	HTTP(80) from alb-sg, SSH(22) from bastion-sg	App accepts from ALB and Bastion only
db-sg	MySQL(3306) from app-sg	DB accepts from App only

Procedure for SG Creation:

1. Go to Security Groups → Create Security Group
2. Give the security group a name and a description Eg: “alb-sg”, “Security group for ALB”
3. Connect to “proj-cas” VPC
4. Add the inbound, Eg: Type: HTTP or SSH, Source: 0.0.0.0/0 or bastion-sg (follow the table above)
5. Leave the outbound rules as default (Allow all)
6. Click “Create Security Group”

Rationale: This follows the principle of least privilege, where each part only accepts traffic from approved sources.

Network ACLs

Purpose: ACLs are subnet-level firewalls which are stateless and hence you must explicitly allow return traffic.

Private-NACL Inbound Rules:

Rule #	Type	Port	Source	Allow/Deny	Purpose
100	HTTP	80	10.0.0.0/16	Allow	Web traffic from VPC
110	SSH	22	10.0.0.0/16	Allow	Admin access from VPC

*	All	All	0.0.0.0/0	Deny	Default Deny

Private-NACL Outbound Rules:

Rule #	Type	Port	Source	Allow/Deny	Purpose
100	HTTP	80	0.0.0.0/16	Allow	Outbound web
110	HTTPS	443	0.0.0.0/16	Allow	Outbound secure web
*	All	All	0.0.0.0/0	Deny	Default Deny

Procedure:

1. Go to Network ACLs → Create network ACL
2. Name: “priv-nacl”, and select “proj-cas” as the VPC
3. Create and select it
4. Edit the inbound and outbound rules
5. Subnet associations → edit and select all 4 private subnets and save

Baston Host Configuration

Purpose: The Bastion host allows a single secure entry point for SSH access to private instances. This access point acts as where the traffic for administrative matters will flow.

Bastion Host Deployment

Procedure:

1. EC2 Console → Launch instances
2. Name: “Bastion-Host”
3. AMI: Amazon Linux 2023 AMI
4. Instance type: t2.micro
5. Key pair: “vockey”

6. Network settings → Edit:
    - VPC: proj-cas
    - Subnet: pub-sub-1a
    - Auto-assign public IP: Enable
    - Security group: Bastion-SG
7. Launch instance

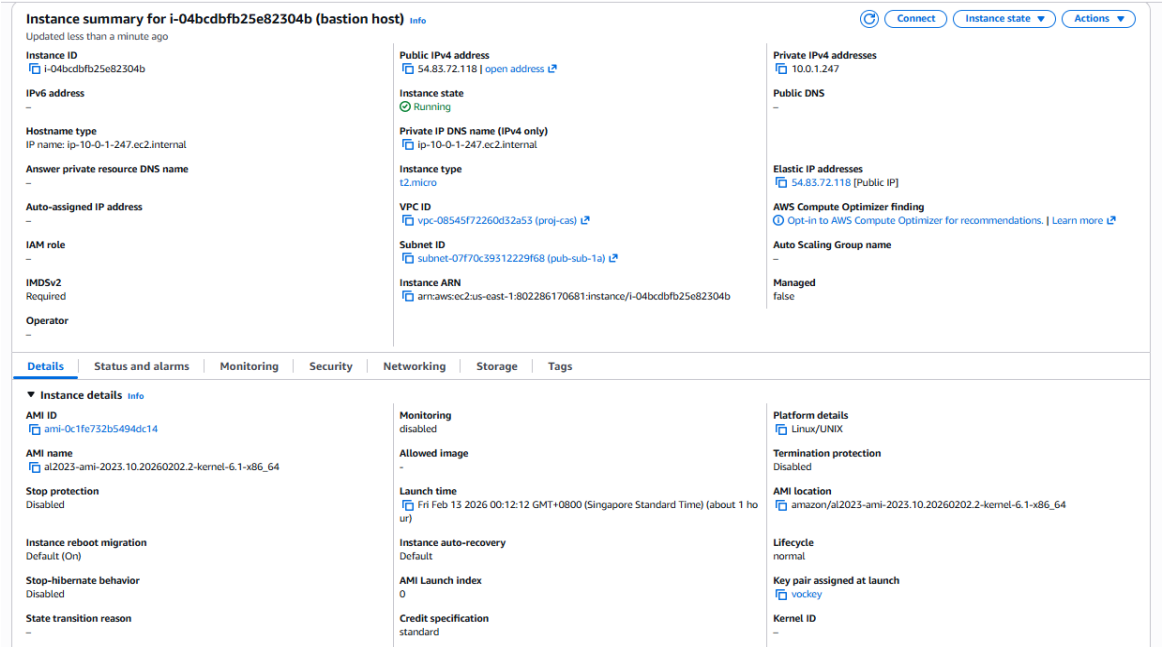


Figure 21: Bastion Host Instance Configurations and Details

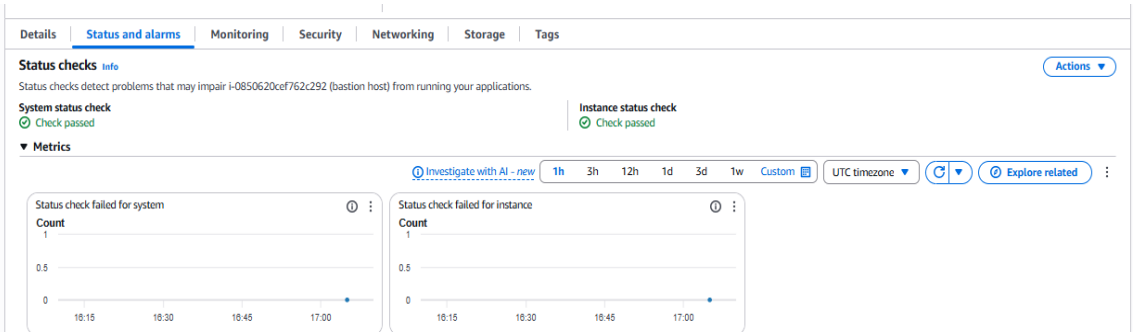


Figure 22: Bastion Host Status and Alarms Tab

Connecting via SSH

From Windows (PuTTY):

1. Open PuTTY, enter: ec2-user@<Bastion-public-IP>
2. Connection → SSH → Auth→ Credentials→Browse for .ppk file
3. Click Open

SSH to Private Instances via Bastion in PuTTY

Procedure:

1. Copy Bastion's public IP address and attach it in PuTTY as "ec2-user@<Bastion-Public-IP>"
2. Open Pageant (PuTTY authentication agent), click "Add Key", select "labuser.ppk" file from its location, and then close the window once the key has been added successfully.
3. In Auth check "allow agent forwarding".
4. In credentials ensure to browse and use the "labuser.ppk" file before clicking open.
5. Once a bird appears, type "ssh ec2-user@<PrivateInstance-Private-IP>
6. If another bird appears, it proves that the connection is successful.

Figure 23: PuTTY showing that Bastion Host Able to Access to Private App Instance

Application Tier

Launch Template

Purpose: Launch Templates help to define configurations for instances that are launched by Auto Scaling.

Configuration:

- Name: app-launch-template

- AMI: Amazon Linux 2023
- Instance Type: t2.micro
- Key Pair: vockey
- Security Group: app-sg

User Data:

```
#!/bin/bash
dnf update -y
dnf install -y httpd mariadb105
systemctl start httpd
systemctl enable httpd

TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-
metadata-token-ttl-seconds: 21600")
INSTANCE_ID=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN"
http://169.254.169.254/latest/meta-data/instance-id)
AZ=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN"
http://169.254.169.254/latest/meta-data/placement/availability-zone)

cat > /var/www/html/index.html << EOF
<!DOCTYPE html>
<html>
<head>
    <title>App Server</title>
    <style>
        body { font-family: Arial; text-align: center; padding: 50px; }
        .box { background: #f0f0f0; padding: 20px; margin: 20px auto; max-width:
400px; border-radius: 10px; }
    </style>
</head>
<body>
    <h1>Application Server</h1>
    <div class="box">
        <p><strong>Instance ID:</strong> $INSTANCE_ID</p>
```

```
        <p><strong>Availability Zone:</strong> $AZ</p>
    </div>
</body>
</html>
EOF
```

Script Explanation:

- Updates system packages
- Installs Apache web server and MySQL client and enables apache to start on boot
- Uses IMDSv2, which are token-based, for metadata access and creates a web page that displays instance ID and AZ for the demonstration

Procedure:

1. EC2 Console → Launch Templates → Create launch template
2. Name: “App-Launch-Template”
3. AMI: Amazon Linux 2023
4. Instance type: t2.micro
5. Key pair: vockey
6. Network settings → Don’t include subnet; Security group: app-sg
7. Advanced details → User data: Paste user data script
8. Create launch template

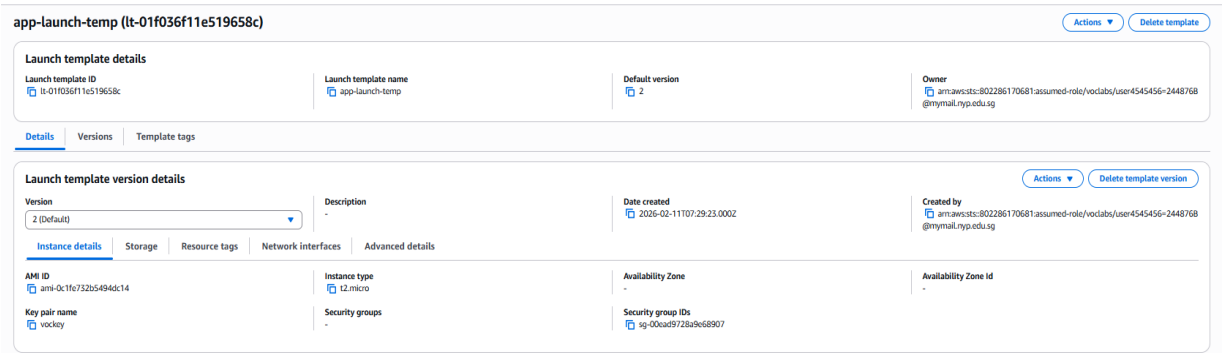


Figure 24: App Launch Template Configurations

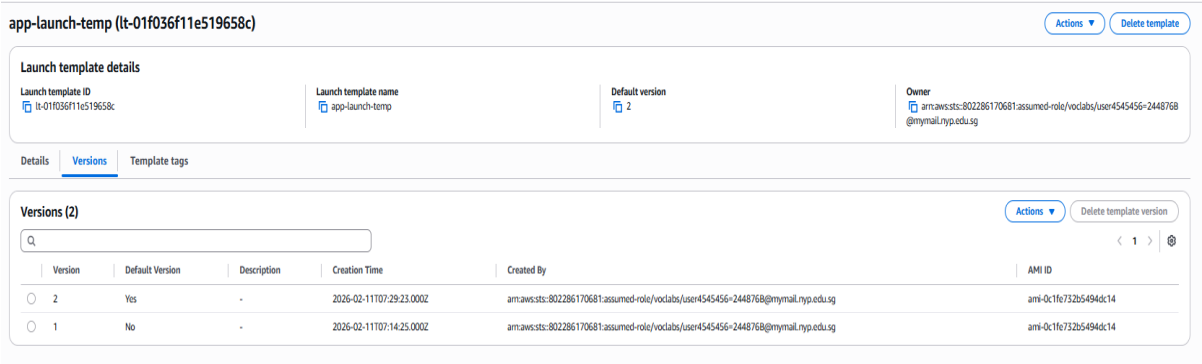


Figure 25: App Launch Template Versions, to showcase ability to update instance configurations

Target Group

Purpose: This directs the load balancer on where to target the traffic to and shows if instances are healthy or unhealthy.

Configuration:

- Name: app-tg
- Target Type: Instances
- Protocol: HTTP, Port: 80
- VPC: proj-cas
- Health check path: /
- Healthy threshold: 2
- Unhealthy threshold: 2
- Timeout: 5 seconds
- Interval: 30 seconds

Procedure:

1. EC2 Console → Target Groups → Create target group
2. Target type: Instances
3. Name: “app-tg”, Protocol: HTTP, Port: 80
4. Choose “proj-cas” as VPC
5. Health check path:/
6. Click Next → Create (do not register targets)

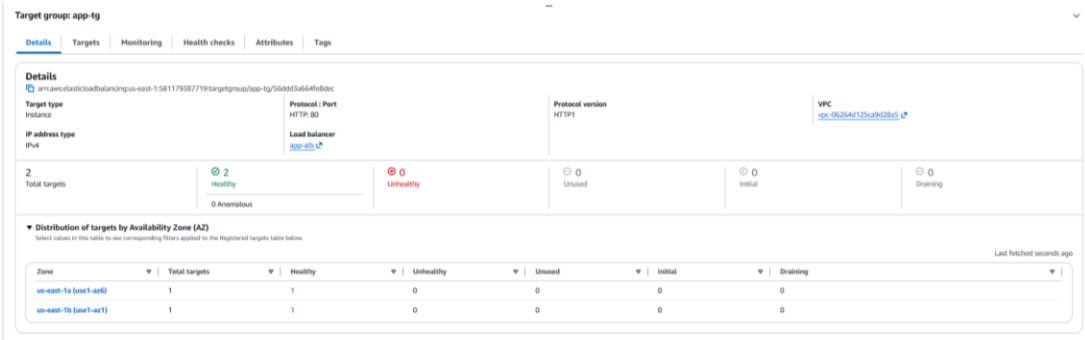


Figure 26: Target Group Details and Configurations

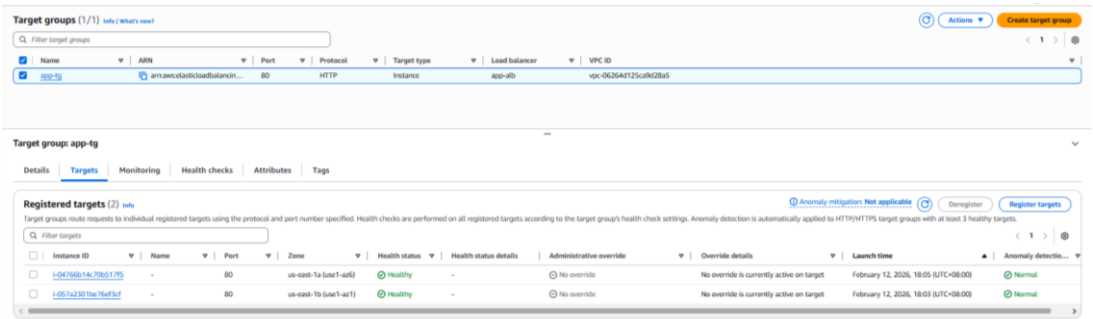


Figure 27: Target Group Targets showing "Healthy" health status

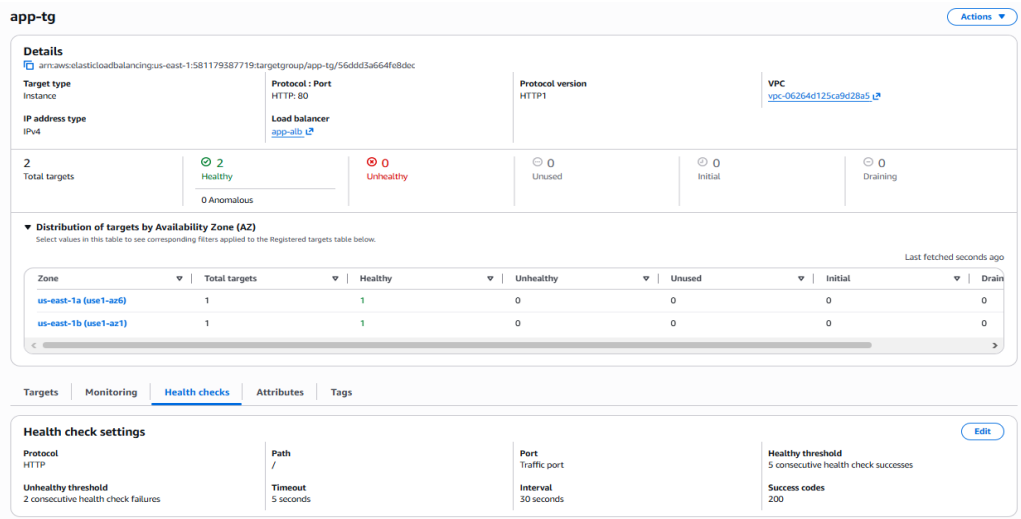


Figure 28: Target Group Health Checks Configurations

Application Load Balancer

Purpose: ALB helps to distribute inbound traffic across the instances in multiple AZs.

Configuration:

- Name: app-alb
- Scheme: Internet-facing
- IP type: IPv4
- VPC: proj-cas
- Subnets: pub-sub-1a, pub-sub-1b

- Security Groups: alb-sg
- Listener: HTTP: 80 → Forward to app-tg

Procedure:

1. EC2 Console → Load Balancers → Create Load Balancer
2. Select Application Load Balancer
3. Name: “app-alb”, Scheme: Internet-facing
4. Network mapping: VPC: proj-cas
5. Select both us-east-1a (pub-sub-1a) and us-east-1b (pub-sub-1b)
6. Security groups: alb-sg
7. Listener: HTTP:80 → Forward to app-tg
8. Create Load Balancer

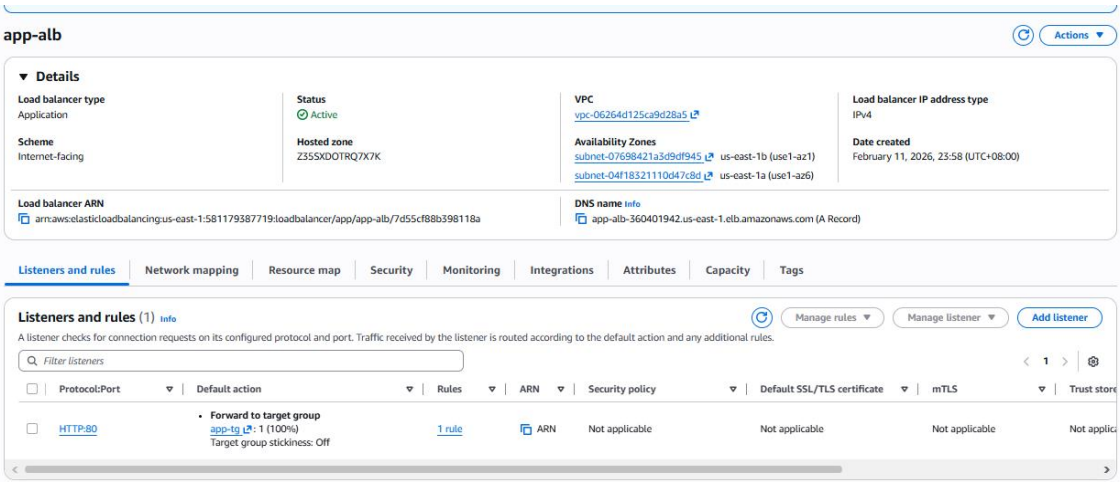


Figure 29: ALB Details and Listeners and Rules

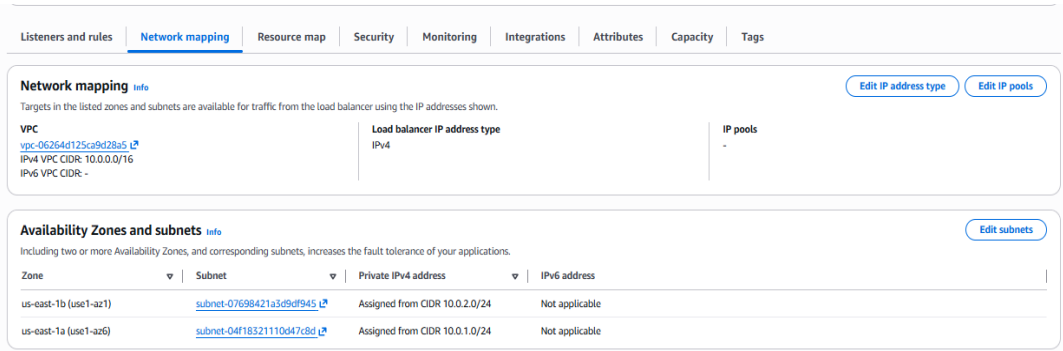


Figure 30: ALB Network Mapping

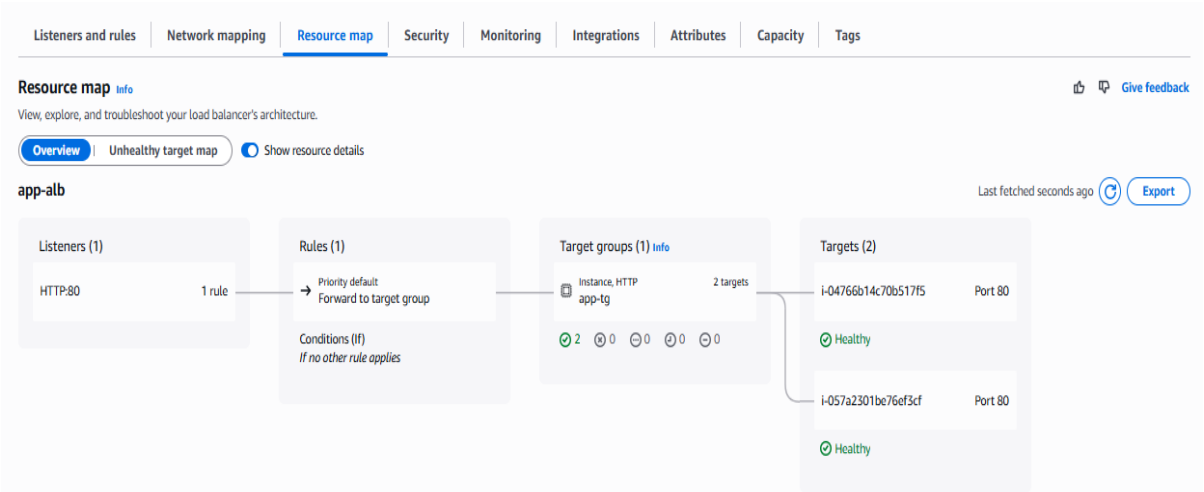


Figure 31: ALB Resource Map

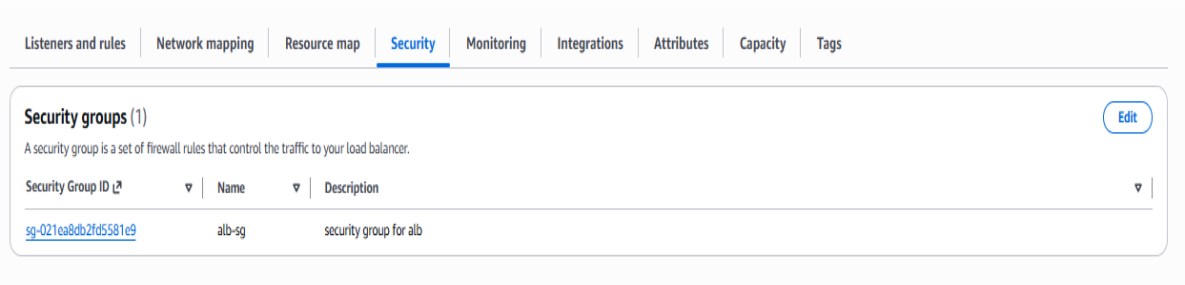


Figure 32: ALB Security Tab

Auto Scaling Group

Purpose: This automatically ensures that the minimum desired number of instances is constantly available. Thus, replacing any unhealthy instances that may occur.

Configuration:

- Name: app-asg
- Launch template: app-launch-template
- VPC: proj-cas
- Subnets: priv-app-sub-1a, priv-app-sub-1b
- Load Balancer: Attached to app-tg
- Desired/Minimum: 2
- Maximum: 4

Procedure:

1. Go to Auto Scaling Groups and create an ASG
2. Name it “app-asg” and choose “app-launch-template”, select “proj-cas” as VPC
3. Subnets: “priv-app-sub-1a” and “priv-app-sub1b”
4. Load balancing: attach to existing load balancer and choose “app-tg”
5. Check “Turn on Elastic Load Balancing Health Checks”

6. Desired: 2, Minimum: 2, Maximum: 4
7. Create Auto Scaling Group

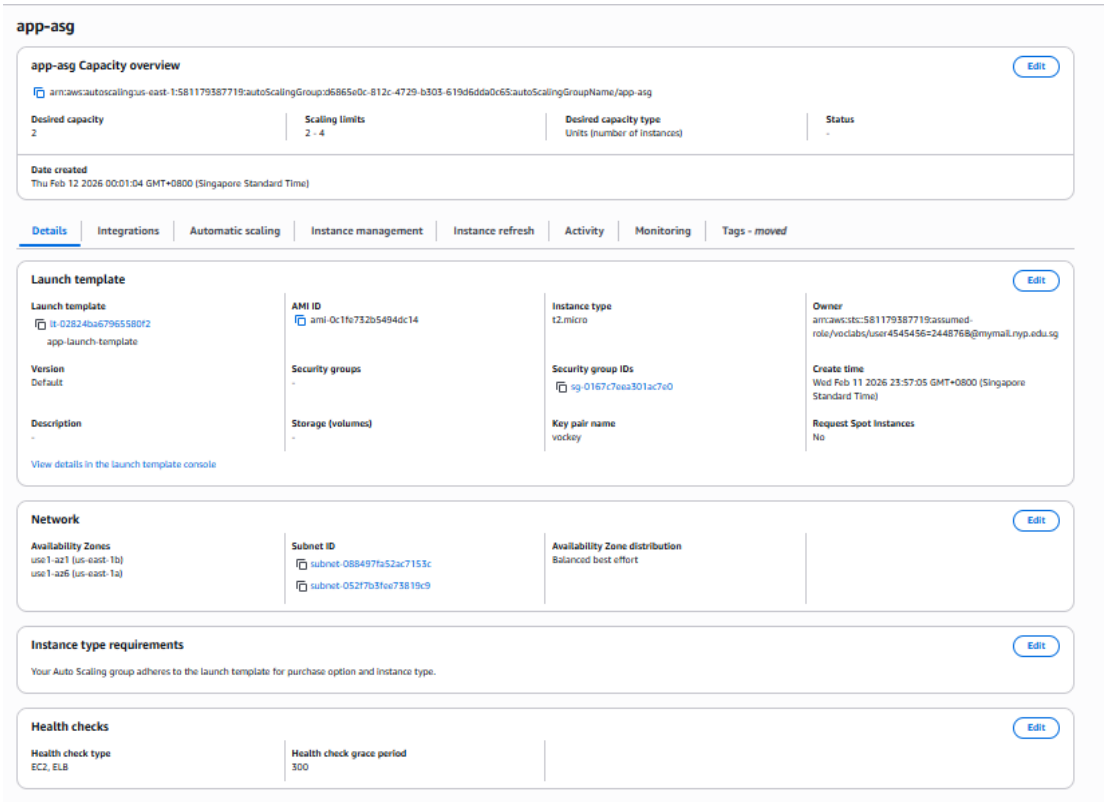


Figure 33: ASG Configurations and Details

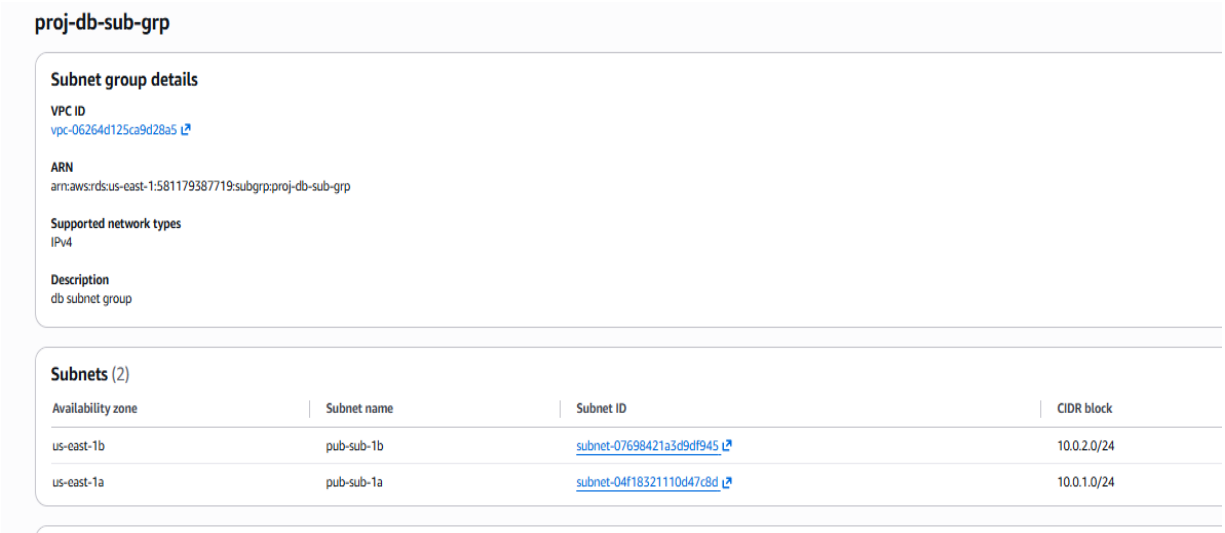
Database Tier - RDS MySQL

DB Subnet Group

Purpose: This helps to define which subnets the RDS can use to deploy.

Procedure:

1. RDS Console → Subnet groups → Create DB subnet group
2. Name: “proj-db-sub-grp”
3. Choose “proj-cas” as VPC
4. Add subnets:
    - AZ: us-east-1a → Priv-db-sub-1a
    - AZ: us-east-1b → Priv-db-sub-1b
5. Create



proj-db-sub-grp

Subnet group details

VPC ID
vpc-06264d125ca9d28a5

ARN
arn:aws:rds:us-east-1:581179387719:subgrp:proj-db-sub-grp

Supported network types
IPv4

Description
db subnet group

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1b	pub-sub-1b	subnet-07698421a3d9df945	10.0.2.0/24
us-east-1a	pub-sub-1a	subnet-04f18321110d47c8d	10.0.1.0/24

Figure 34: Database Subnet Group Details

RDS MySQL Instance

Configuration:

- Engine: MySQL 8.0
- Template: Dev/Test (for Multi-AZ)
- DB Instance ID: project-mysql-db
- Master Username: admin
- Instance Class: db.t3.micro (burstable)
- Storage: 20GB gp3
- Multi-AZ: Yes (standby in different AZ)
- VPC: proj-cas
- Subnet Group: proj-db-sub-grp
- Public Access: No
- Security Group: db-sg

Procedure:

1. RDS Console → Databases → Create database
2. Choose Standard Create and MySQL the default version as engine
3. Template: Dev/Test and name the db instance identifier as “proj-mysql-db”
4. Master username: admin

5. Master password: hello123
6. Instance Class: db t3.micro (Choose burstable)
7. Storage: 20GB, gp3, disable autoscaling
8. Multi-AZ deployment: Create a standby instance
9. VPC: “proj-cas”
10. Subnet Group: proj-db-sub-grp
11. Public access: no
12. Security Group: db-sg and remember to remove default-sg
13. Click create database and wait for it to have “Available” status which can take up to 15 minutes.

Rationale: Deploying the RDS in multiple AZs will provide us with automatic failover, so if the primary AZ fails, the standby is promoted to be the primary within 60 seconds. Public access should be disabled for security purposes and only allow app-sg to connect to the RDS through private IP.

Testing Database Connectivity

From Bastion Host:

```
sudo dnf install mariadb105 -y
mysql -h <RDS-ENDPOINT> -u admin -p
```

Expected Results: It should not be successful as the database should only be accessible through private IP through the App Instances.

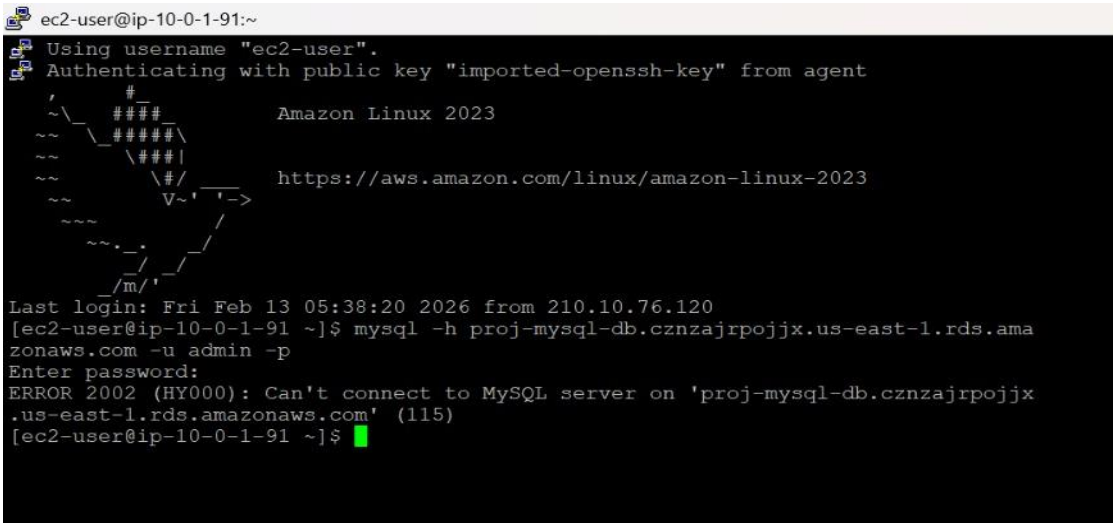


Figure 35: Bastion (Public IP address) trying to access into RDS but Failed

From App Instance (via Bastion):

```
Mysql -h <RDS-ENDPOINT> -u admin -p
```

Expected Results: Successful connection should be seen.

```
[ec2-user@ip-10-0-1-91 ~]$ ssh ec2-user@10.0.10.166
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-10-166 ~]$ mysql -h proj-mysql-db.cznzajrpojjx.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 282
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

Figure 36: Accessing RDS through App Instance (Private IP Address)

S3 Storage Configuration

Requirement: Application instances must download from S3 without storing access keys in code.

Method: Access a public S3 bucket via HTTPS URL using curl (no AWS credentials required).

Upload a file (index.html) into S3 Bucket and uniquely name it “s3-bucket-egl211”.

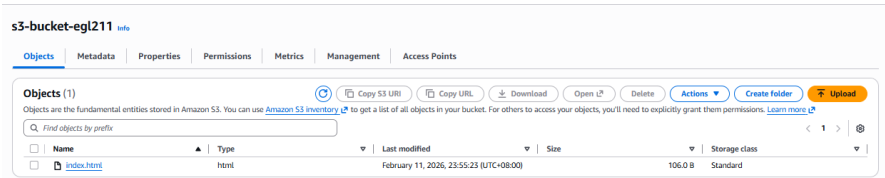


Figure 37: S3 Bucket Objects Uploaded (index.html)

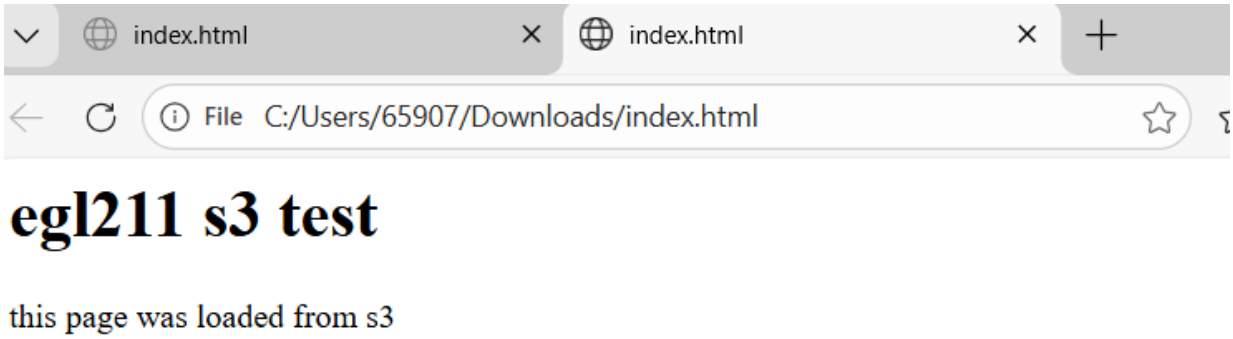


Figure 38: Contents of index.html file inserted into S3 Bucket

S3 Bucket Policy:

```
{
        "Sid": "DenyInsecureTransport",
        "Effect": "Deny",
        "Principal": "",
        "Action": "s3:",
        "Resource": [
            "arn:aws:s3::: s3-bucket-egl211",
            "arn:aws:s3::: s3-bucket-egl211/"
        ],
        "Condition": {
            "Bool": {
                "aws:SecureTransport": "false"
            }
        }
    },
    {
        "Sid": "DenyDeleteObject",
        "Effect": "Deny",
        "Principal": "",
        "Action": "s3:DeleteObject",
        "Resource": "arn:aws:s3::: s3-bucket-egl211/"
    },
```

```
        {
            "Sid": "AllowPublicUpload",
            "Effect": "Allow",
            "Principal": "",
            "Action": "s3:PutObject",
            "Resource": "arn:aws:s3::: s3-bucket-egl211/*"
        }
    ]
}
```

1. DenyInsecureTransport

This statement denies all S3 actions if the request is not made over HTTPS.

Explanation:

- “Effect”: "Deny" blocks access.
- "Action": "s3:*" applies to all S3 operations.
- "Condition": { "aws:SecureTransport": "false" } means any HTTP (non-HTTPS) request is rejected.
- This ensures encrypted communication between client and S3.

2. DenyDeleteObject

This statement prevents objects in the bucket from being deleted.

Explanation:

- "Effect": "Deny" overrides any allow permissions.
- "Action": "s3:DeleteObject" blocks delete operations.
- Protects stored files from accidental or malicious removal.

3. AllowPublicUpload

This statement allows users to upload objects to the bucket.

Explanation:

- "Effect": "Allow" grants permission.
- "Action": "s3:PutObject" allows uploading files.
- "Resource": "arn:aws:s3:::egl211-cas-bucket/*" applies to all objects in the bucket.
- This enables public upload functionality.

Production Recommendation

In production with proper permissions:

1. Create IAM Role with S3 read access
2. Attach role to EC2 instances via Launch Template
3. Use aws s3 cp commands (credentials managed automatically by role)

EBS Snapshots

Purpose: This provides backups to ensure business continuity and disaster recovery.

Creating a Snapshot:

1. Go to Instances in the EC2 console
2. Choose one of the App instances and go to the storage tab
3. Press the volume ID link and select the volume
4. Click Actions and Create Snapshot
5. Add tag: Name = “app-snaoshot”
6. Create Snapshot

Use Cases:

- This helps to recover data if the instance was corrupted
- Allows us to launch a new instance from the snapshot
- Allows for cross-region disaster recovery
- It is compliant and meets audit requirements

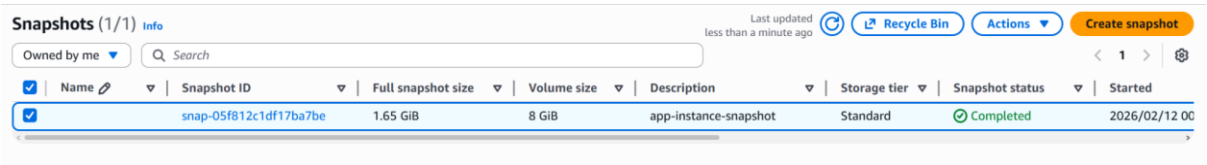


Figure 39: Snapshot made

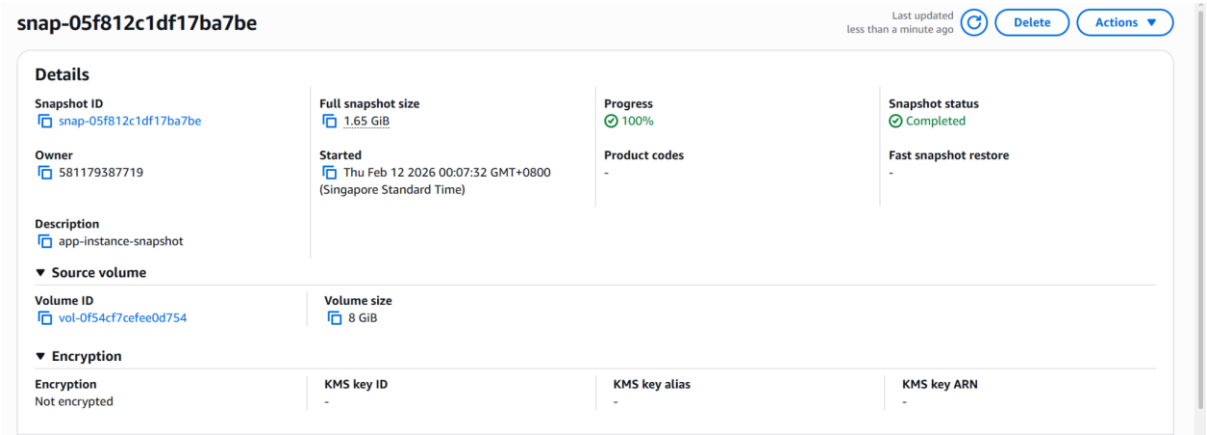


Figure 40: Details of the Snapshot that was created

CloudWatch Monitoring

Purpose: CloudWatch alarms provide proactive monitoring and alerting for infrastructure health

ALB Unhealthy Host Alarm

Configuration:

- Metric: UnHealthyHostCount
- Namespace: AWS/ApplicationELB
- Condition: Greater than 0
- Period: 1 minute

Procedure:

1. Go to CloudWatch Console → Alarms → Create Alarm
2. Select metric → ApplicationELB → Per AppELB, per TG metrics
3. Select UnHealthyHostCount for App-ALB and App-TG
4. Statistic: Average, Period: 1 minute
5. Condition: Greater than 0
6. Configure notification → Alarm Name: “alb-unhealthy-host-alarm”
7. Create alarm

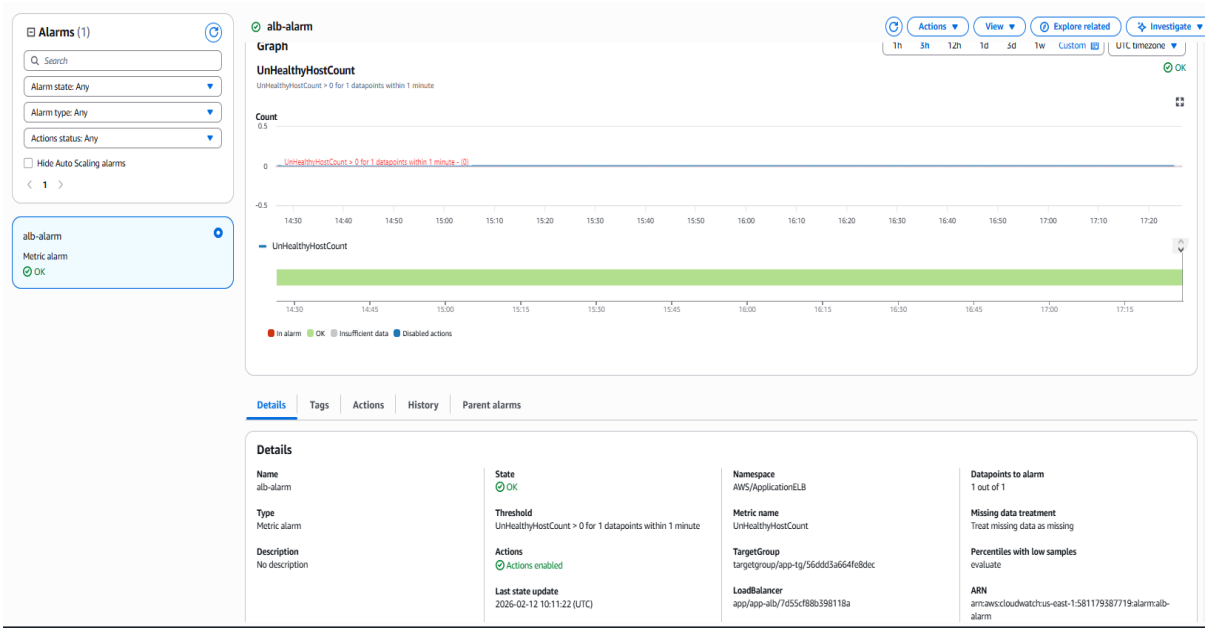


Figure 41: CloudWatch Alarm Configurations

Testing and Validation

Web Application Access

Test: Access web page via ALLB DNS name

Procedure:

- Open browser → http://<ALB-DNS-NAME>

Expected Results: Web page displays Instance ID and Availability Zone

Verification: Refresh multiple times - Instance ID changes (proves load balancing)

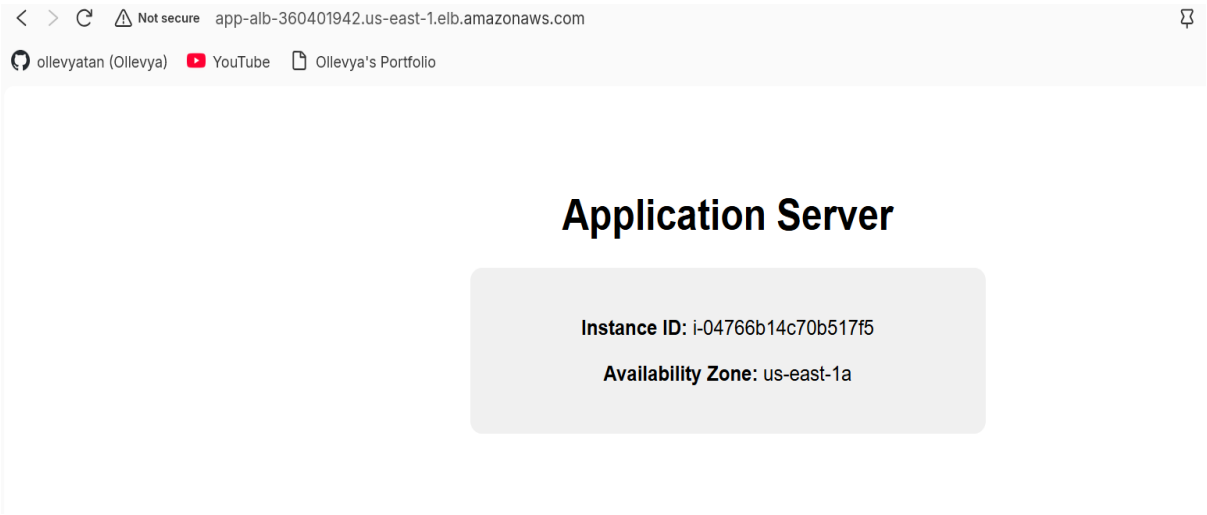


Figure 42: Web server showing what instance is running and which availability zone it is in

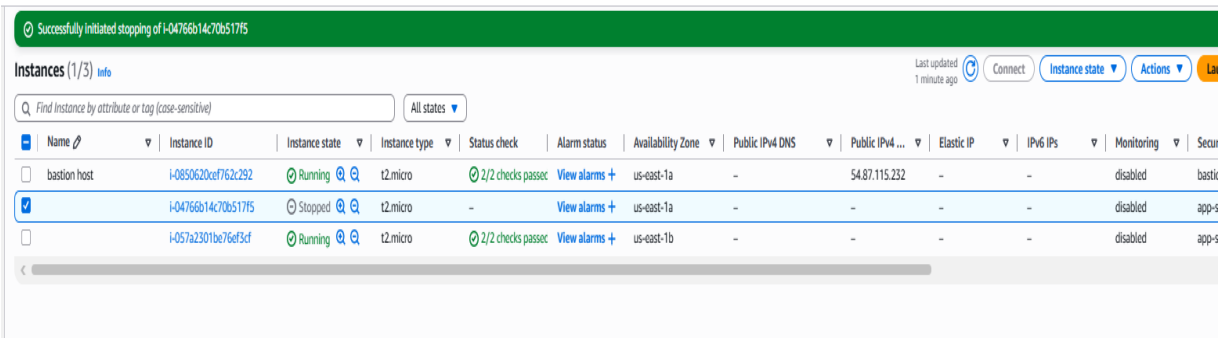


Figure 43: Shows that when one instance is stopped

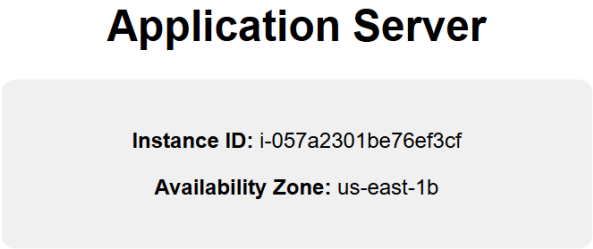


Figure 44: Web server after instance was stopped, shows that load balancer is working because it shows a different instance in a different AZ

High Availability Test

Test: Stop one instance, verify traffic continues

Procedure:

1. Note current Instance IDs from web page
2. EC2 Console → Stop one App-Instance
3. Refresh web page

Expected Result: Page still loads, shows only remaining instance

Auto Scaling Self-Healing

Test: Terminate instance, verify ASG replaces it Procedure:

1. Terminate 1 App Instance
2. Wait 2 - 3 minutes
3. Check EC2 instances and ASG activity tab

Expected Result: New Instance launched automatically

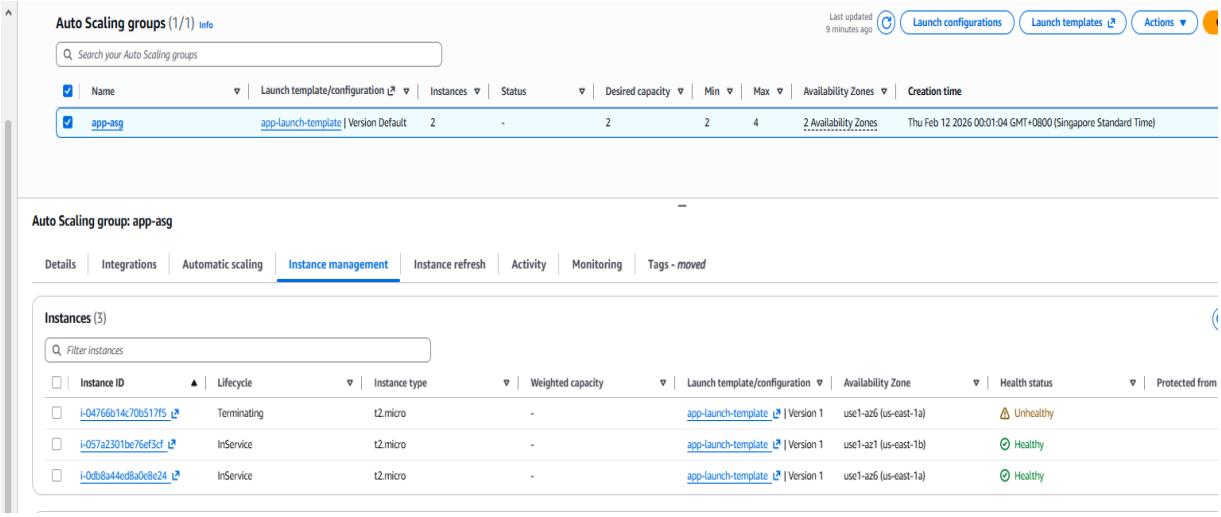


Figure 45: One instance was terminated and it shows an "Unhealthy" health status

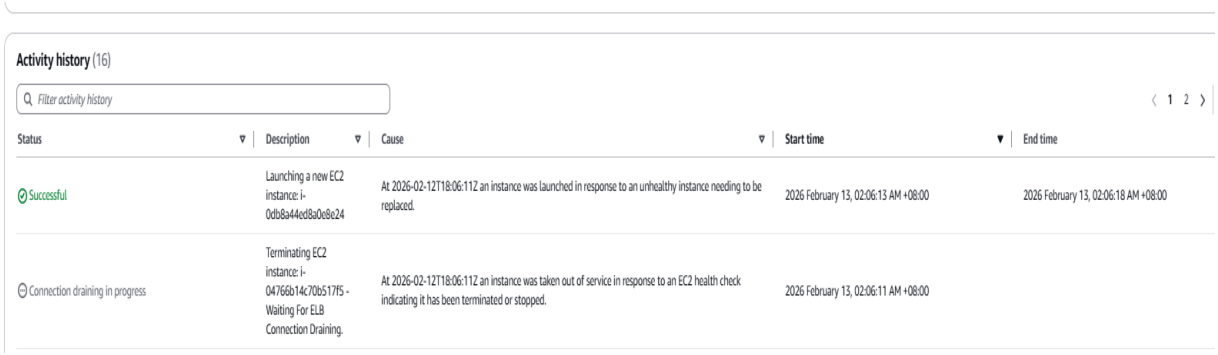


Figure 46: The Activity History shows that a new instance was launched successfully

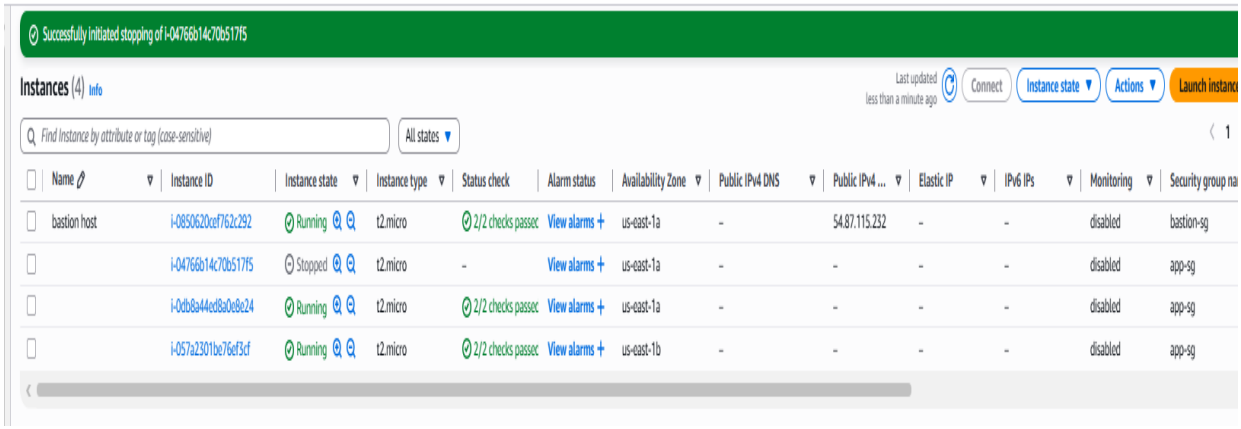


Figure 47: This shows the old instance that was terminated and the new instance running, always showing that there's constantly 2 app instances

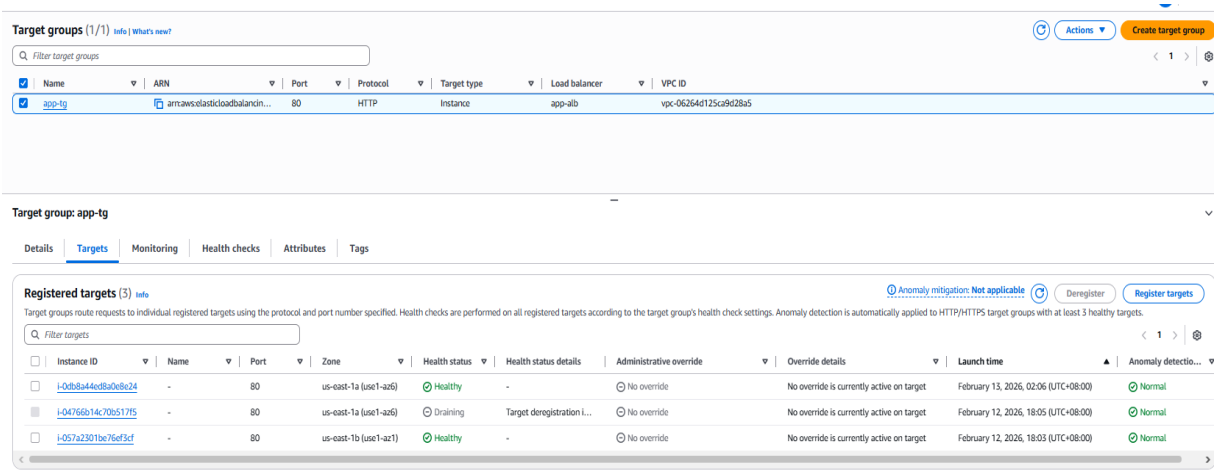


Figure 48: Target group shows 2 healthy and running instances after draining the terminated instance

Bastion SSH Access

Test: SSH from local → Bastion → App instance

Procedure: execute SSH chain as documented in 5.3.5

Expected Result: Successful connection to private instance via Bastion only

Database Connectivity

Test: Connect to RDS from app instance

Procedure: From app instance in PuTTY:

```
mysql -h <RDS-ENDPOINT> -u admin -p
```

Expected Result: Successful MySQL connection

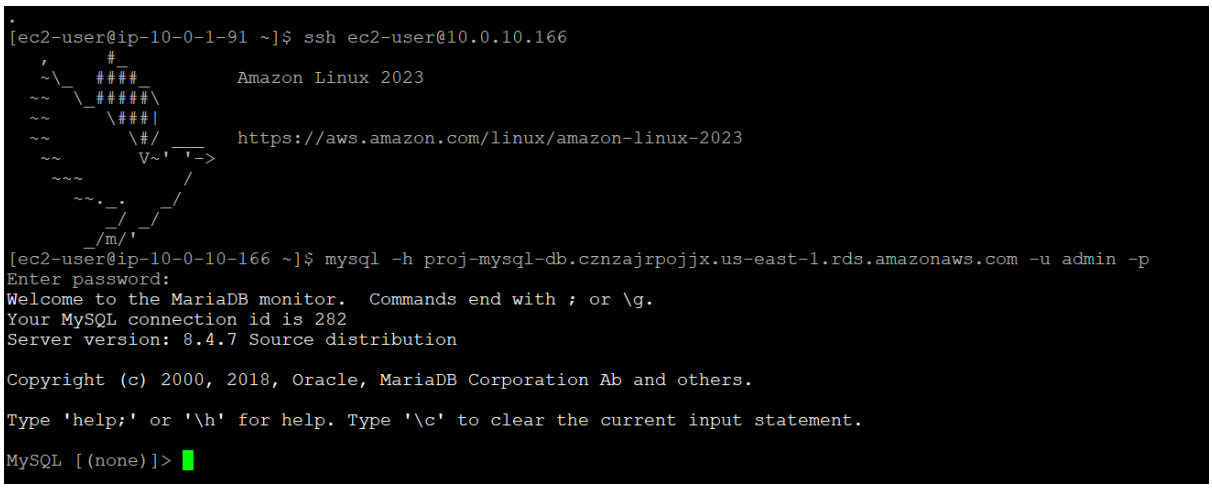


Figure 49: Shows Successful MySQL connection to RDS from Private app instance through PuTTY

Database Not Publicly Accessible

Test: Attempt a connection from Bastion instance in PuTTY

Procedure: From local:

```
mysql -h <RDS-ENDPOINT> -u admin -p
```

Expected Result: Connection timeout which proves that the DB is not accessible through public IP.

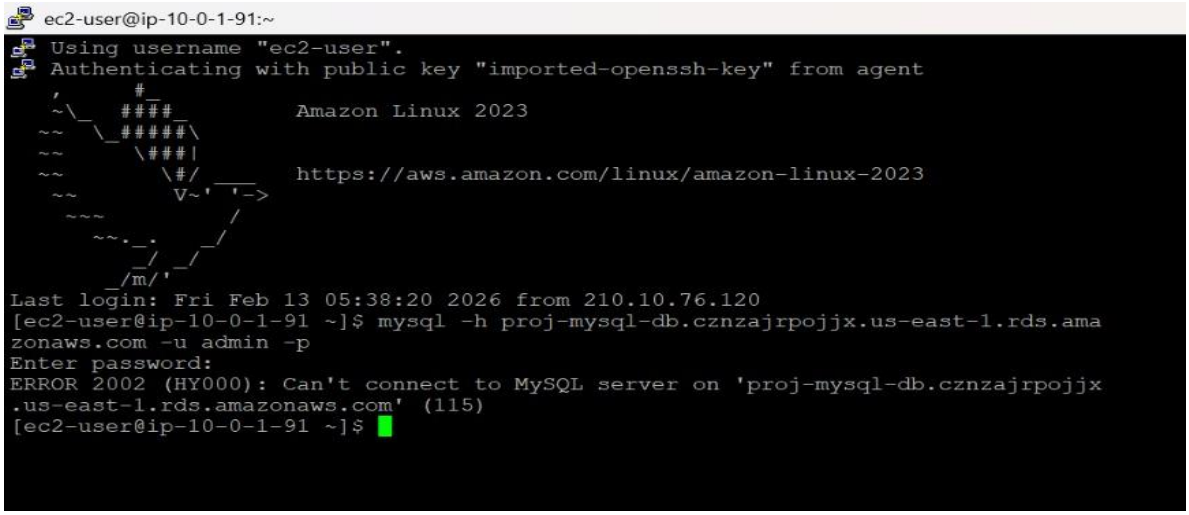


Figure 50: Attempted connection to RDS from Bastion failed as Bastion uses public IP

Private Instance Internet Access

Test: Verify that NAT Gateway is working

Procedure: From app instance

```
curl -I https://google.com
```

Expected Result: HTTP response received -proves private instances can reach the internet

S3 Testing

To see if S3 can be downloaded without any stored credentials in the instance.

Test Command (from App Instance in PuTTY):

```
curl -O https://bucket-proj-egl211.s3.amazonaws.com/index.html
cat index.html
```

What this proves:

- App instance in private subnet can reach the internet (NAT Gateway working)
- Can download from S3 without any credentials stored on the instance
- Demonstrates the architecture pattern (in production, use IAM roles)

Figure 51: Curl S3 Bucket in Private App instance

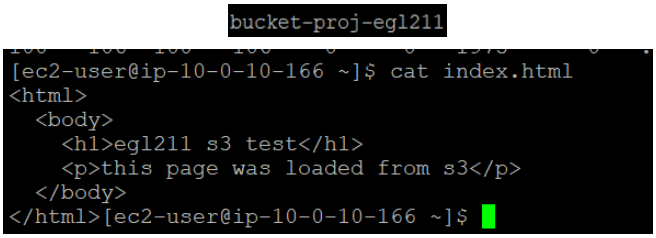


Figure 52: Reading index.html file and displaying its contents

In Private EC2 instance, try:

aws s3 ls s3://bucket-proj-egl211

Expected output: Unable to load credentials

It shows that no IAM role is attached and no access keys are configured.

Problem & Challenges

IAM Role Creation Restrictions

Problems: AWS lab account did not permit IAM role creation.

Impact: Could not implement IAM role for S3 access (best practice).

Solution: Used public S3 bucket with HTTP URL access to demonstrate connectivity. In production, IAM roles should be used for secure, credential-free access.

EC2 Metadata Services (IMDSv2)

Problem: Initial webpage showed blank Instance ID and Availability Zone.

Cause: Amazon Linux 2023 uses IMDSv2 by default, requiring token-based authentication for metadata requests.

Solution: Updated user data script to:

1. First obtain a token using PUT request
2. Use token in subsequent metadata API calls

Original (failed):

```
INSTANCE_ID=$(curl -s http://169.254.169.254/latest/meta-data/instance-id)
```

Fixed:

```
TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" -H-aws-ec2-metadata-token-ttl-seconds: 21600")
INSTANCE_ID=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/instance-id)
```

Bastion Host IP Change

Problem: Could not SSH to Bastion after stopping / starting instance

Cause: EC2 instances get new public IP when restarted unless using Elastic IP.

Solution: Check EC2 console for the new public IP after every restart. We attach the Elastic IP to Bastion for persistent addressing which is for production purposes.

Individual Reflections

Yadanar’s Reflection

Implementing and validating the Database Tier helped me better understand how to balance High Availability with security. Deploying an AWS RDS Multi-AZ instance was important because it provides automatic synchronous replication to a standby instance and promotes it quickly if the primary fails. This showed me how cloud architecture supports resilience beyond basic deployment. At the same time, disabling public access reinforced the importance of keeping the database fully private and accessible only through controlled internal layers. I also learned the significance of the RDS DNS endpoint, which remains unchanged during failover, allowing the application to stay connected without requiring configuration changes.

From a security perspective, I applied the Principle of Least Privilege to ensure that traffic flowed strictly through the intended architecture path. Testing connectivity was an important learning point. When attempting to connect from the Bastion Host, the connection correctly failed, confirming that the database was not directly accessible. However, connecting from the App Instance via the Bastion was successful, validating that the trusted access path was working as designed. Overall, this phase strengthened my understanding of how managed databases can provide both reliability and security within a well-architected cloud environment.

Ollevya’s Reflection

During the implementation of the bastion host architecture, I gained a deeper understanding of how AWS networking and security controls work together to reduce attack surface. Initially, I underestimated the impact of security group source restrictions and user data scripts on connectivity and system behaviour. Through troubleshooting SSH timeouts and MySQL connection delays, I learned how package

management processes dnf and NAT Gateway performance can affect instance availability.

This task reinforced the importance of the principle of least privilege, particularly in designing security groups that allow access only from trusted sources rather than open IP ranges. I also developed a clearer distinction between administrative access (bastion host) and application access, which is critical in real-world cloud environments.

Overall, this exercise improved my confidence in designing secure, highly available AWS architectures and highlighted the value of deliberate planning before deployment to avoid unnecessary rework.

Min’s Reflection

Working on the load balancing part helped me understand more clearly how AWS keeps applications stable and available. By setting up the Application Load Balancer together with the Target Group and Auto Scaling Group, I learned how incoming traffic can be shared across multiple servers instead of depending on just one. When I tested it by refreshing the ALB page and seeing different instance IDs appear, it made the concept feel very real and practical. I also learned that health check settings are very important, because if they are not configured properly, instances can be marked unhealthy and won’t receive traffic. Troubleshooting this taught me how routing and monitoring work together. Overall, this task gave me a better hands-on understanding of high availability and scalability in a simple and practical way.

Zilds’s Reflection

During this project, I took ownership of the data protection strategy and the technical integration of our tiered architecture. Implementing Requirement 4.4 through EBS Snapshots taught me the practical application of Point-in-Time recovery; by capturing a point-in-time image of our application volume, I ensured that we had a reliable recovery point to mitigate risks like data corruption or accidental deletion. This highlighted the importance of "Business Continuity" in a professional Associate Engineer role, moving beyond simple deployment into long-term infrastructure stability.

Beyond my specific tasks, I acted as a technical lead for the group's troubleshooting efforts. I guided team members through the complexities of Step 10 (RDS Deployment) and verified the multi-step SSH connection paths required to reach our private resources via the Bastion Host. A significant challenge was ensuring everyone understood the "jump" logic—moving from the local terminal to the Bastion, then to the App or DB layers—which solidified my understanding of secure administrative access.

To further safeguard our progress, I maintained a backup project configuration (referenced in our VPC and security checklists) to ensure we had a functional baseline in case of configuration drift. Overall, this phase strengthened my ability to not only build cloud components but also to debug the networking "glue" that allows them to function securely as a cohesive system.

Angelin’s Reflection

During this project, I learned how to manage teams and gained a deeper understanding of Cloud in general. The difficulty of working in a team whereby the software could not be shared really posed as a challenge to all of us. We had to constantly be aware of what each other’s tasks were and whether they were completed. Because there are so many ways to deploy systems in AWS, the struggle for me was that everyone’s methods were right and it caused a lot of confusion.

This just proves the importance of communication between team members and the importance of updating without needing to ask. Apart from that, I really gained a lot of understanding on implementation of VPCs and subnets and how they should be routed. As well as EC2 instances and why some may need to have private IPs only and not accessible through the internet. This project has taught me a lot of critical thinking skills, and my brain was tested to its maximum capacity when going through troubleshooting. Troubleshooting problems like being unable to connect to PuTTY happened often. Whether there was a problem with security group inbound rules or a problem with the app instance user data, it was not an easy feat to troubleshoot them. But nonetheless, a lot of valuable lessons were learned, and I look forward to doing the next project.

Brandon’s Reflection

During this task, I learned that setting up cloud storage is not just creating a bucket and upload file only. At first, I thought it was very simple, but when doing the bucket policy and testing access, I realised security settings are very important and

sometimes can block myself also if too strict. I also faced some confusion between public and private download, and I learned how preassigned URL works to give temporary access without saving access keys inside EC2.

I also understand more about least privilege and why we should only allow the actions we really need. Sometimes the AWS lab limitations made things harder, especially IAM role and permissions, so I had to find alternative ways to make it work. Overall I learned how different AWS services connect and how small mistakes in policy can affect the whole system. This project helped me better understand practical cloud security and troubleshooting step by step.

Recommendations

All recommendations were researched through reading through AWS technical documentation.

Security Enhancements (AWS, 2024)

1. **AWS WAF:** Implement Web Application Firewall on ALB to protect against SQL injection, XSS, and other web exploits.

2. **VPC Flow Logs:** Enable flow logs for network traffic monitoring and security analysis (requires IAM permissions).

3. **AWS CloudTrail:** Enable API activity logging for auditing and compliance.

4. **Restrict Bastion SSH:** Change Bastion-SG inbound rule from 0.0.0.0/0 to specific administrator IP addresses.

5. **Secrets Manager:** Use AWS Secrets Manager for database credentials instead of hardcoded passwords.

Cost Optimization (AWS,2024)

1. **Reserved Instances:** For predictable workloads, purchase Reserved Instances for significant cost savings.

2. **Right-sizing:** Monitor instance utilization and adjust instance types accordingly.

3. **NAT Gateway Alternatives:** Consider NAT instances for development environments to reduce costs.

Operational Excellence (AWS, 2024)

1. **Infrastructure as Code:** Use CloudFormation or Terraform for reproducible, version-controlled deployments.
2. **CI/CD Pipelines:** Implement automated deployment pipelines for application updates.
3. **Comprehensive Monitoring:** Implement CloudWatch dashboards and additional alarms for proactive monitoring.
4. **Backup Strategy:** Implement automated RDS snapshots and cross-region backup replication.

Conclusions

In this project we successfully implemented a 3-tier architecture on AWS which is high available and meets all the requirements that were needed.

- Creating a custom VPC with 6 subnets across 2 AZs with proper routing towards IGW and NAT GW.
- Using ALB to distribute traffic across ASG-managed instances and includes health checks.
- Deployed a multi-AZ RDS using MySQL that is not publicly accessible by only allowing app-sg to access port 3306 in inbound rules.
- Created Bastion Host to ensure secure SSH access to private instances.
- Deployed EBS Snapshots to ensure business continuity

- Able to download S3 without stored credentials from private instances
- Deployed in-depth defence systems through security groups, network ACLs, and CloudWatch monitoring.

Through this project, we were able to deepen our understanding of AWS networking concepts and implement high availability across multiple AZs. We also learned about including least privilege as one of the best security practices. Implementing auto scaling to improve resilience and cost optimisation and have managed database services for efficiency. We learned that with a good architecture, cloud infrastructure can automatically recover from failures, be scaled to meet demands and maintain security without needing human interventions.

References

Rance, K. (2024). *The rise of cloud Computing: Transforming IT infrastructure*. 83Zero. https://www.83zero.com/blogs/the-rise-of-cloud-computing-transforming-it-infrastructure/

Gridelli, S. (2025, 29 January). *What is an RFC 1918 Address?* Netbeez. https://netbeez.net/blog/rfc1918/

Senarath, U. S. (2025, November 4). *Understanding subnets in Amazon VPC: The Foundation of AWS Networking.* Medium. https://medium.com/@udarasenarath/understanding-subnets-in-amazon-vpc-the-foundation-of-aws-networking-a3e4c252c6cc

Amazon Web Services. (2024). *Security group rules*. https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html

Amazon Web Services. (2024). *AWS CloudTrail user guide*. https://docs.aws.amazon.com/awscloudtrail/latest/userguide/

Amazon Web Services. (2024). *VPC flow logs*. https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html

Amazon Web Services. (2024). *AWS WAF developer guide*. https://docs.aws.amazon.com/waf/

Amazon Web Services. (2024). *AWS Secrets Manager user guide*. https://docs.aws.amazon.com/secretsmanager/latest/userguide

Amazon Web Services. (2024). *Reserved instances*. https://aws.amazon.com/ec2/pricing/reserved-instances/

Amazon Web Services. (2024). *Cost optimization pillar – AWS Well-Architected framework*. https://docs.aws.amazon.com/wellarchitected/latest/cost-optimization-pillar/

Amazon Web Services. (2024). *NAT gateways and NAT instances*. https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat.html

Amazon Web Services. (2024). *AWS CloudFormation user guide*. https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/

Amazon Web Services. (2024). *What is CI/CD?* https://aws.amazon.com/devops/continuous-delivery/

Amazon Web Services. (2024). *Amazon CloudWatch user guide*. https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/

Amazon Web Services. (2024). *Backing up and restoring an Amazon RDS DB instance*. https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_CommonTasks.BackupRestore.htm

Appendices

```
MySQL [(none)]> exit
Bye
[ec2-user@ip-10-0-10-51 ~]$ curl "https://bucket-proj-egl211.s3.us-east-1.amazon
aws.com/index.html?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=ASIA3VS73OI
47SG3KBHF%2F20260212%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20260212T073502Z
&X-Amz-Expires=300&X-Amz-SignedHeaders=host&X-Amz-Security-Token=IQoJb3JpZ2luX2V
jEAgaCXVzLWVhc3QtMSJIMEYCIQCYtFzL5ypdWKNMoOp4Jg3ieWFBKBja6OnSGdqWGfBsUwIhAKbLMkB
%2B5RDsdZ5EsUK56I2Eqf81t7rsbxUHj6sQ2IcOKpADCNH%2F%2F%2F%2F%2F%2F%2F%2F%2F%2FwEQA
RoMODAyMjg2MTcwNjgxIgypj33qF3CWre6Rl1Mq5AKQOTDVR%2BptOdrrUMh3KIY8rA3p0ilYeubvtQH
%2FQnf%2BLP9jhDG9MXbGWqWCHAAUTBWVaURDkewO1nH6SbHIaoHuzYyYwJRrMgqsj7JetmBC8JwTGzd
hUIFXmQjbO36A7XnY5%2BYpIt56brId3FZDNvasUVXcWVN41z4zbgb1jxPT%2BYGDcJiHwn1AKX0TAkI
RUfgCQkD0XL8%2Fgvrzkxbxy LXIv2r7%2FX9LXxswt9gP3CdNB%2Fv701Md%2B0JNzeb%2F69n4gOzbK
9XhY0ELfsvXO5fS0xSonBxo73VqRIvuqwX0Bru6O8AB3sYyZlcJKH%2B33or8YJcoaSZcZI7jMfvhHQf
f6IImlYW7Z3EpugYfszn8ojiRNzUzq0DAMBfDYwPRKEv4N%2Bp3Yxv2IM0pXbbDpm6%2BXlBUmzNmQNB
dtS8UqCafOFnLh3FA3Ybwkt4FAm7iFi9hsEn37vy0m8dwVG28ijvhIvNpP2sW%2BzDH4bXMBjqGAiqO%
2Bf%2FsNjjMQEO7VhquscB2EUHytSeQnbr7E7UZVTBXIOA55wRaKC1onOLnHNghOk4JAS97af%2FmEmj
tKFV1q7%2FXMMeKKJkeT9NwUSbs9YEMdkGPfVQNf1WsDJ1BHh7OIexZ3dCTqzvRIKo4Kv6JTNP7NULhn
GxeMIK851HGJutsQRKrKUbLijlb%2B8hk1%2B%2BemfWdHNCjzlRbyFt%2Fzs9GojO1f11GTagZLzEDr
hiLjGJaZf7zc7nVx70uQmrAbvvTzdjwiD6SPlSFa90CpnI4eL0o43zkQ%2FBWthdDr7nMq2BWBG%2Fcq
wN8b1uLNm4LZklQ84ZQqYaffsapgJ51puFTaK18YRRRoTc%3D&X-Amz-Signature=d42a7c3c7ce332
83056b9f4b6a7704ac83a1e705feff44c61df8ac4ead989f7f" -o index.html
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
```

Figure 53: Accessing S3 Bucket through CloudShell Link

sg-079025d16150be9e5 - alb-sg2

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0672fb7494cb1e411	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
-	sgr-021af7c47b58dc435	IPv4	HTTP	TCP	80	0.0.0.0/0	-

Figure 54: Inbound rules for alb-sg

sg-079025d16150be9e5 - alb-sg2

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination	Description
-	sgr-02556282152c27f58	IPv4	All traffic	All	All	0.0.0.0/0	-

Figure 55:Outbound rule for alb-sg

sg-00ead9728a9e68907 - app-sg

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-052b211d98ffe74f9	-	SSH	TCP	22	sg-079014cf1b09b5c64 / bastion-sg	-
-	sgr-0f46086755119db7b	-	HTTP	TCP	80	sg-079025d16150be9e5 / alb-sg2	-

Figure 56: Inbound rules for app-sg

sg-00ead9728a9e68907 - app-sg

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination	Description
-	sgr-00fa49499db9e02bc	IPv4	All traffic	All	All	0.0.0.0/0	-

Figure 57:Outbound rules for app-sg

sg-079014cf1b09b5c64 - bastion-sg

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (1)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0abeb9e7418cfa052	IPv4	SSH	TCP	22	0.0.0.0/0	-

Figure 58: Inbound rules for bastion-sg

Figure 59: Outbound Rules for bastion-sg

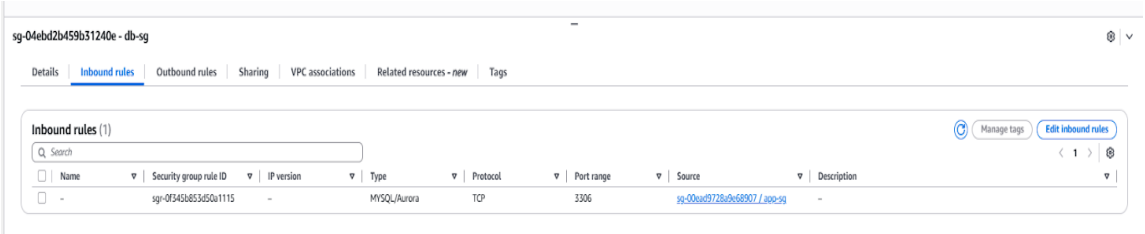


Figure 60: Inbound rules for db-sg

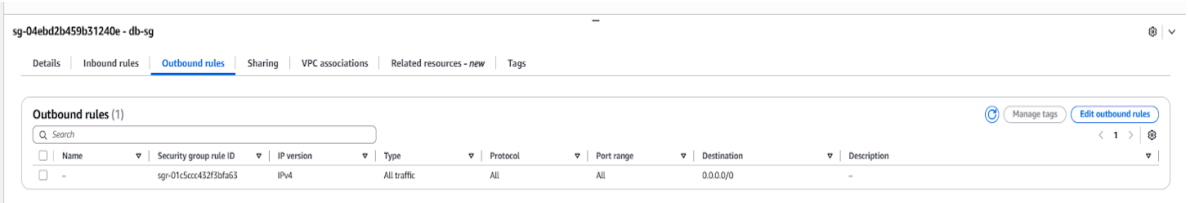


Figure 61: Outbound rules for db-sg

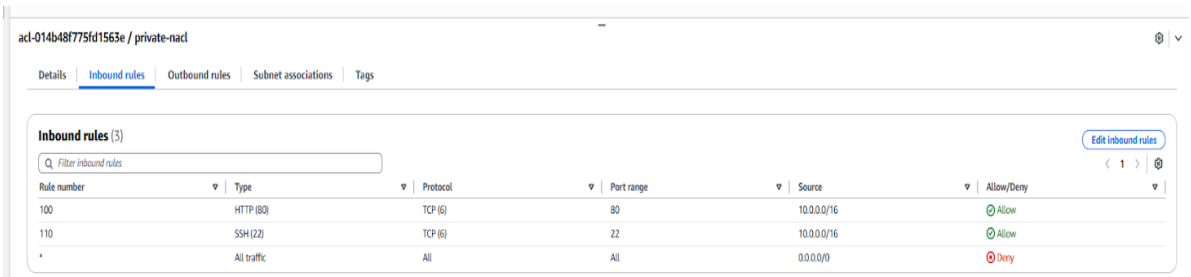


Figure 62: Inbound rules for private-nacl

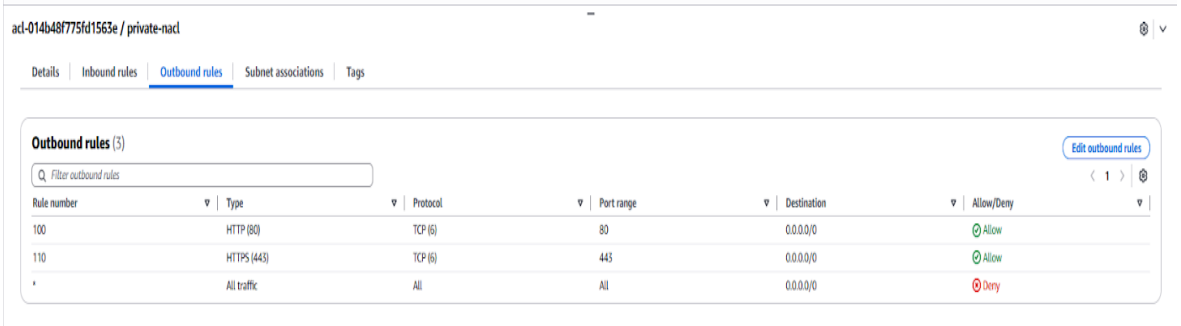


Figure 63: Outbound rules for private-nacl

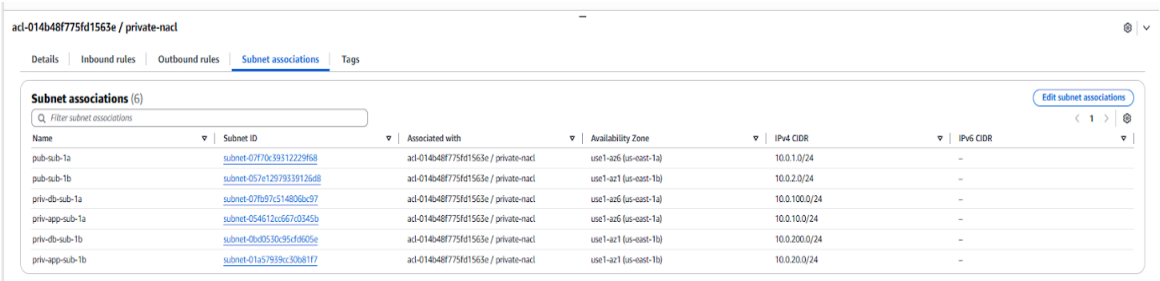


Figure 64: subnet associations for private-nacl

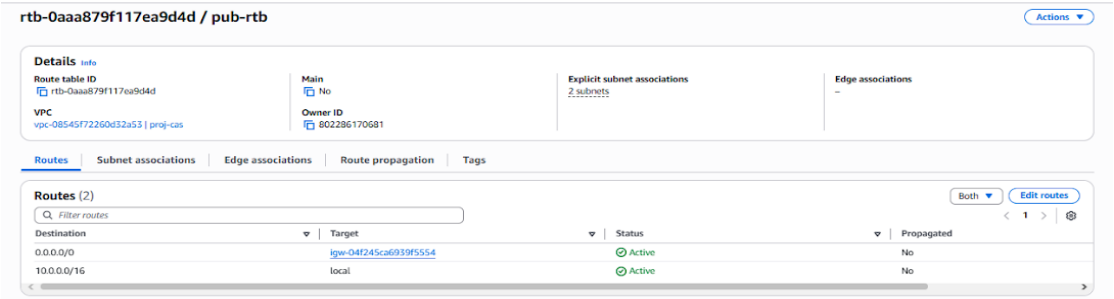


Figure 65: public route table routes

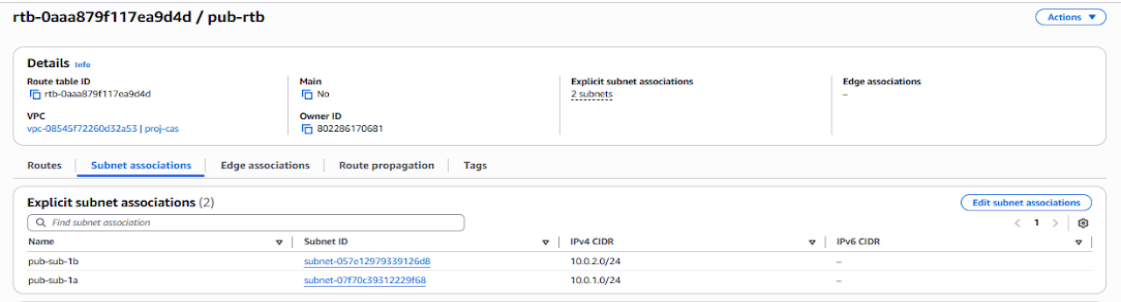


Figure 66: Public Route table subnet associations

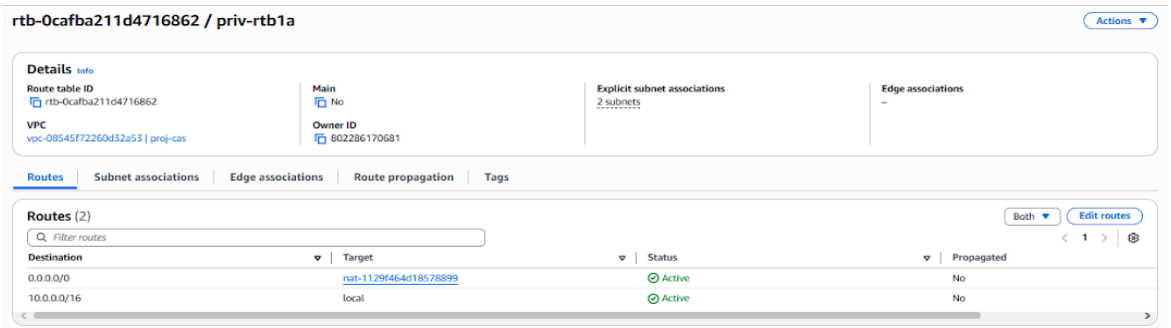


Figure 67: Private route table us-east-1a routes

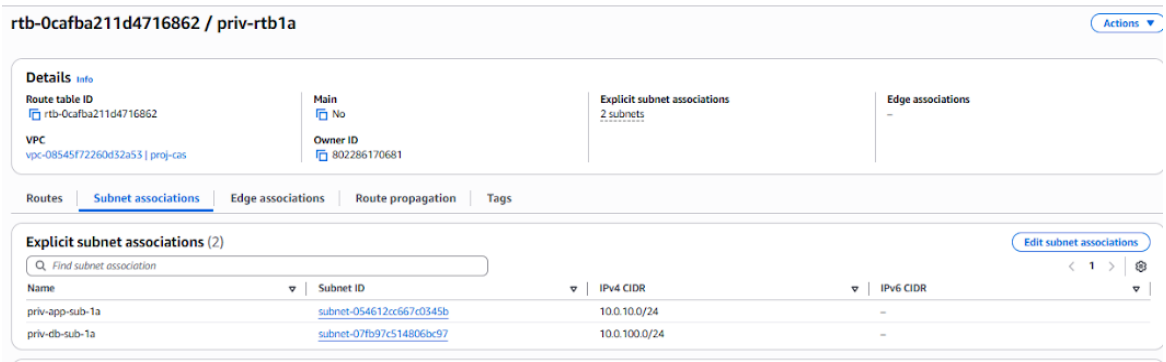


Figure 68: Private route table us-east-1a subnet associations

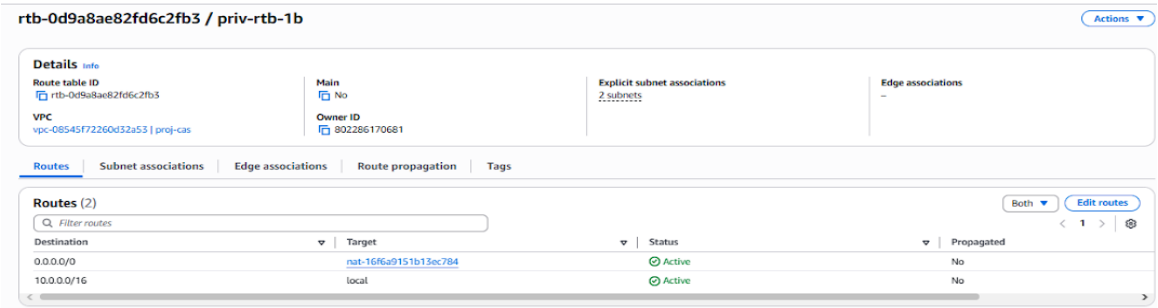


Figure 69: Private route table us-east-1b routes

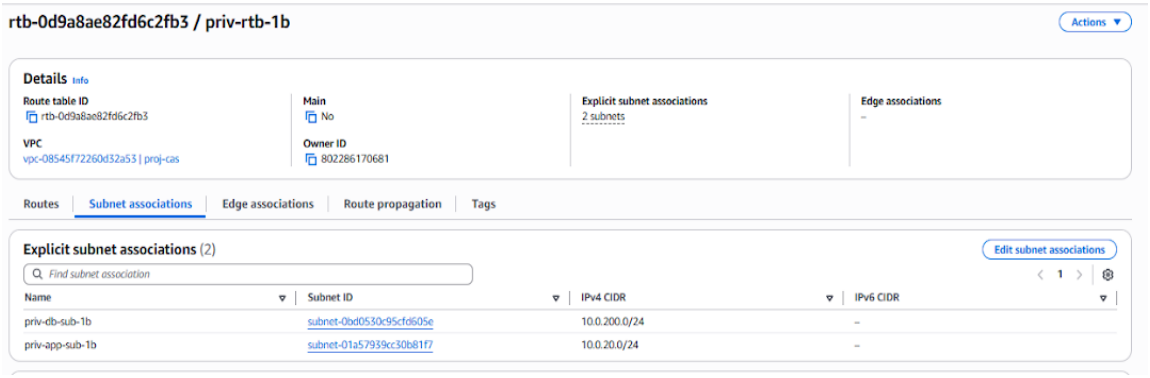


Figure 70: private route table use-east-1b subnet associations